

FINITE-USE GEOMETRY AND LEARNING OF ORDERED QUANTUM MEASUREMENTS

QIAN QI

ABSTRACT. We study finite-use discrimination of ordered quantum measurements with classical output and no residual device system. A normalized effect-tuple covariance has a complete support-coordinate kernel. For every fixed two-sided C^2 measurement curve with nonzero first derivative H , the Hermitian matrix $\Gamma_E(H) = \frac{i}{2} \sum_j [\text{supp } E_j, H_j]$ determines the finite-use scale. The distance is comparable to $\min\{1, \sqrt{N}|t|\}$ when this matrix belongs to the support span, and to $\min\{1, N|t|\}$ otherwise. Constants and the two-sided interval depend on the fixed curve. Second-order rank openings are allowed. An analytic horizontal surrogate and a corrected logical channel give direct finite-angle bounds; certified control errors have a separate explicit accumulation bound. The exponents and error-correction principle are established in channel metrology; the formulation here uses effect coordinates without assuming a smooth exact Kraus representation at the boundary. The arbitrary-pair covariance theorem remains an upper bound. Transported-weight regularization obeys classical data processing. Balanced-interior description and learning results retain their fixed-outcome-count optimal orders, with growing-outcome dependence separately displayed. Exact first-order decisions and affine legalization are polynomial on rational input; general recovery and dictionary synthesis are not inferred from them. The principal support and discrimination proofs are self-contained in this article; complete binary auxiliary material is preserved in a current supplement.

1. INTRODUCTION

A measurement with a classical output may retain a quantum advantage when its input is correlated with a reference. The question considered here is quantitative: how well can two ordered measurements be separated with at most N uses, and which features of their effects determine the dependence on N ? The answer must accommodate both strictly positive effects and singular measurements. Positivity of each effect alone is not the relevant constraint: the effects sum to the identity, and their boundary geometry is coupled by this equation.

1.1. The model and the principal results. An ordered k -outcome measurement on $\mathbb{C}^d$ is a tuple $E_j \succeq 0$ with $\sum_j E_j = I_d$. One call is the channel

$$\mathcal{M}_E(\rho) = \sum_{j=1}^k \text{tr}(E_j \rho) |j\rangle \langle j|.$$

The input is consumed and the output is only the classical label; no residual device system is available. A tester may prepare inputs entangled with its own reference, retain quantum memory, and choose subsequent controls using previous labels. It uses the same control rule under both hypotheses. Its public stopping time is bounded by N . We define $D_N(E, F)$ to be the supremum of the unhalved trace distance between its two final states. Thus $0 \leq D_N \leq 2$. Nonadaptive testers may have entangled inputs and retained references; “nonadaptive” does not mean “product input.”

Date: 5 October 2026.

2020 Mathematics Subject Classification. Primary 81P15, 81P47; Secondary 94A17, 81P70.

Our complete-body upper bound uses a positive, concave covariance $\mathcal{C}_E$ on zero-sum Hermitian tuples. Its factorization through a horizontal projection gives a finite-use bound valid through changes of rank and zero effects. With $M = (E + F)/2$ and $H = F - E$,

$$(1.1) \quad D_N(E, F) \leq \min \left\{ 2, 2\sqrt{k+1} \sqrt{N \langle H, (\mathcal{C}_M + N^{-1} \text{Id})^{-1} H \rangle} \right\}.$$

This is a global upper certificate, not a claimed formula for the optimal adaptive distance. The normalization includes the exact binary Sylvester restriction of Theorem 2.1.

The new support description determines its entire kernel. If $P_j = \text{supp } E_j$ and $\mathcal{W}_E = \sum_j P_j \mathcal{H}_d P_j$, then Theorem 4.1 parametrizes all null directions and proves

$$\dim \ker(\mathcal{C}_E|_{\mathcal{T}}) = d^2 - \dim \mathcal{W}_E + \sum_j (d - \text{rank } E_j)^2.$$

This distinction between support span and the mere effect span is essential for effects of rank greater than one. In particular, a nonprojective measurement can have a nontrivial covariance kernel; the earlier projective characterization concerns a zero operator, not one null direction.

This kernel has a finite-use interpretation on every fixed unitary orbit $E_j(t) = e^{itG} E_j e^{-itG}$. Apart from the stationary case, Theorem 5.1 proves, uniformly in every integer $N \geq 1$ and all sufficiently small $|t|$,

$$(1.2) \quad D_N(E, E(t)) \asymp_{E,G} \begin{cases} \min\{1, \sqrt{N}|t|\}, & G \in \mathcal{W}_E, \\ \min\{1, N|t|\}, & G \notin \mathcal{W}_E. \end{cases}$$

The second lower bound uses a corrected logical qubit and the explicit finite-angle estimate (5.9). The support span is precisely the Hermitian Kraus-product span of this measurement channel. The metrological criterion is credited to Zhou–Jiang [40]; the present formulation identifies it with the covariance kernel and proves the finite-pair trace estimates directly. This is an orbit theorem on the complete body, not a full-boundary volume or minimax theorem.

We also replace a coordinate-fixed ridge by a public scalar-measurement covariance. Transporting its probability vector through the same output channel gives a regularized data-processing inequality and exact invariance under reversible label splitting (Theorem 8.1). The original fixed Euclidean ridge is not asserted to have this property.

1.2. Nonunitary curves and certified controls. The support description also gives a local classification without a unitary-orbit assumption. For a fixed two-sided C^2 measurement curve with $H = E'(0) \neq 0$, set

$$\Gamma_E(H) = \frac{i}{2} \sum_j [\text{supp } E_j, H_j].$$

Theorem 6.2 proves

$$D_N(E(0), E(t)) \asymp_{E(\cdot)} \begin{cases} \min\{1, \sqrt{N}|t|\}, & \Gamma_E(H) \in \mathcal{W}_E, \\ \min\{1, N|t|\}, & \Gamma_E(H) \notin \mathcal{W}_E. \end{cases}$$

This holds for every finite N on a two-sided interval depending on the fixed curve. The supports may change at second order. An analytic horizontal surrogate supplies a $\sqrt{N}|t| + O(Nt^2)$ upper in the range branch; a support-complement code supplies a logical channel with an explicit $O(t^2)$ remainder in the other branch. Both are finite inequalities rather than inferences from asymptotic Fisher information. The metrological exponents and correction mechanism remain attributed to [40, 41]; the new formulation works directly with C^2 effects even when an exact Kraus family is not assumed differentiable at a rank opening.

Theorem 7.1 bounds the loss from uniformly certified errors in preparation, encoding, recovery and readout. The allowed per-cycle error scales with $|t|\Delta$, where Δ is the logical gap. This is

a conditional stability guarantee for known-pair controls, not an executed implementation or a gate-synthesis complexity theorem. The constants and interval may degenerate with support eigenvalues, tangent size, curvature and the support residual. Zero first derivative does not imply stationarity for a general curve. The global arbitrary-pair covariance bound remains one-sided.

1.3. Learning and descriptions on balanced interiors. The operational consequences are complemented by exact description and common learning laws on the balanced interior $I_d/(2k) \preceq E_j \preceq 3I_d/(2k)$. Its real affine dimension is $p = (k - 1)d^2$. Theorems 9.2 and 9.4 give

$$\log_2 \mathcal{C}_{N,k}(\delta) = \frac{p}{2} \log_2 N + p \log_2(1/\delta) + O(p \log(k + 1)), \quad 0 < \delta < 1/128.$$

The common learner has the joint order

$$M_{d,k}^*(N, \delta, \eta) = \Theta_k \left(N \delta^{-2} [d^2 + d \log(1/\eta)] \right)$$

for fixed $k, d, N \geq 1$, $0 < \delta \leq 2^{-24}$ and $0 < \eta \leq 1/8$. The displayed constructive upper has a k^3 factor; the lower does not match growing k . Theorem 11.1 supplies polynomial, search-free statistical legalization, while the entropy-optimal public dictionary remains a separate finite, potentially exhaustive construction. The lower theorem permits arbitrary coherent adaptive training, whereas the upper uses fresh acquisitions and collective processing of completed outputs. Pair-dependent discrimination witnesses do not serve as a common learning design.

1.4. Resource conventions and reading order. Unknown-device calls, future-use horizon, transmitted index length, public dictionary reconstruction, classical arithmetic, trusted-control synthesis, and physical execution are distinct resources. A query bound here holds on every record, with a defined legal fallback. Public dictionaries and their parameters are independent of the target; target-dependent private advice would count as transmitted information. Exact rational certificate evaluation and balanced legalization have polynomial bit complexity. Neither statement asserts polynomial synthesis of an arbitrary collective readout or of the whole entropy-optimal dictionary. The orbit recovery is an ideal-control comparison construction, not an executed common learner.

Sections 2–8 develop the global upper, support kernel, orbit classification and transported ridge. Sections 9–11 give the balanced-interior learning and coding consequences. All binary auxiliary proofs used by this article are supplied in the separate *Binary Foundations and Auxiliary Proofs* supplement. Cross-document theorem references carry the prefix S and are resolved from its current source, not from a historical PDF. The primary article and supplement together are a self-contained proof package; the independent structural paper is not a premise of any measurement theorem.

1.5. Logical prerequisites of the primary article. The covariance factorization, variance identity and full-body path upper are proved in Section 2; the finite Bernoulli lower is proved in Section 3. Together with the interface above, these suffice for the support and finite-discrimination results of Sections 4–7. The independently complete structural paper is never a premise.

Primary result	Required auxiliary result
Support kernel, orbit and smooth-curve classification	None from the supplement: all probability, covariance and correction premises are proved in the primary.
Transported regularization	Primary covariance Jensen identity and variational duality.
Balanced-interior common learning	Supplement Theorem S15.3 and Corollary S12.4; the former imports the credited binary tomography primitive.
Noisy-projector comparison	Supplement Theorem S4.1 for the specified binary pair. It is not a premise of the smooth-curve theorem.
Other binary boundary, coding and control proofs	Preserved in the current supplement; not premises of the new support or smooth-curve classification.

2. A COVARIANCE PRINCIPLE ON THE CLOSED MEASUREMENT BODY

Let $\mathcal{P}_{d,k}$ denote the complete body of ordered k -outcome measurements on $\mathbb{C}^d$: its elements are Hermitian tuples $E = (E_1, \dots, E_k)$ with $E_j \succeq 0$ and $\sum_j E_j = I_d$. Here $d \geq 1$, $k \geq 2$, and $N \geq 1$ are integers. Zero effects, changing ranks, and noncommuting supports are allowed throughout this section. The operational distance D_N uses unhalved trace norm, retained references, common adaptive control, and at most N calls with bounded public stopping. The device consumes its input and returns only its ordered classical label.

The normalized tangent space and its inner product are

$$\mathcal{T}_{d,k} = \{H_j = H_j^* : \sum_j H_j = 0\}, \quad \langle H, K \rangle = \sum_j \text{tr}(H_j K_j).$$

All operator inequalities below on tuple spaces refer to this real Hilbert structure, unless stated otherwise. For any Hermitian tuple K , not necessarily of zero sum, put

$$(2.1) \quad S_E(K) = \sum_j E_j K_j,$$

$$(2.2) \quad (\mathcal{C}_E K)_j = \frac{1}{2}(E_j K_j + K_j E_j - E_j S_E(K) - S_E(K)^* E_j).$$

The matrix $S_E(K)$ need not be Hermitian. Keeping its adjoint in (2.2) is essential. The operator $\mathcal{C}_E$ is polynomial in the effects, annihilates constant tuples, and has range in $\mathcal{T}_{d,k}$. In particular its restriction to $\mathcal{T}_{d,k}$ is well defined.

For $\tau = N^{-1}$ set

$$(2.3) \quad g_{N,E}(H)^2 = N \langle H, (\mathcal{C}_E + \tau \text{Id})^{-1} H \rangle,$$

$$(2.4) \quad \mathcal{Q}_N(E, F) = g_{N, (E+F)/2}(F - E).$$

The inverse acts on $\mathcal{T}_{d,k}$. Positivity proved below makes it nonsingular even on the spectral boundary. We do not identify $\mathcal{Q}_N$ with an operational distance or assume a triangle inequality for it.

Theorem 2.1 (Closed-body adaptive bound). *For every $E, F \in \mathcal{P}_{d,k}$,*

$$(2.5) \quad D_N(E, F) \leq \min\{2, 2\sqrt{k+1} \mathcal{Q}_N(E, F)\}.$$

More generally, every continuously differentiable path $E(t)$ in $\mathcal{P}_{d,k}$ satisfies

$$(2.6) \quad D_N(E(0), E(1)) \leq \sqrt{k+1} \int_0^1 g_{N, E(t)}(E'(t)) dt.$$

These statements are uniform over all ranks and support patterns. For $k = 2$, $E = (A, I_d - A)$ and $H = (B, -B)$, the restriction is exactly

$$(2.7) \quad (\mathcal{C}_E H)_1 = A(I_d - A)B + BA(I_d - A), \quad \mathcal{Q}_N(E, F)^2 = 2Q_N(A, F_1)^2,$$

where Q_N is the binary comparison modulus in (S4.4).

Thus the binary variance operator has a normalized-tuple extension, not an independently regularized list of effects. The theorem gives a global upper certificate. The two-sided balanced-interior classification and its entropy theorem remain Theorems 9.1 and 9.2; a matching general lower bound for (2.4) is a separate question.

2.1. Positive covariance and horizontal decomposition.

Lemma 2.2 (Covariance identities). *On all Hermitian tuples, $\mathcal{C}_E$ is self-adjoint and positive semidefinite, and*

$$\begin{aligned} \langle K, \mathcal{C}_E K \rangle &= \sum_j \text{tr}(E_j K_j^2) - \|S_E(K)\|_{\text{HS}}^2 \\ (2.8) \quad &= \sum_j \|E_j^{1/2}(K_j - S_E(K))\|_{\text{HS}}^2. \end{aligned}$$

For $E_t = (1-t)E + tF$, $0 \leq t \leq 1$,

$$(2.9) \quad \langle K, [\mathcal{C}_{E_t} - (1-t)\mathcal{C}_E - t\mathcal{C}_F]K \rangle = t(1-t)\|S_F(K) - S_E(K)\|_{\text{HS}}^2.$$

The kernel consists precisely of tuples satisfying $E_j^{1/2}(K_j - S_E(K)) = 0$ for every j . Moreover $\mathcal{C}_E = 0$ on $\mathcal{T}_{d,k}$ if and only if the E_j are mutually orthogonal projections, with zero projections permitted.

Proof. Write $A_j = E_j^{1/2}$ and let A be the vertical stack of the A_j . Then $A^*A = I_d$, so $P = I - AA^*$ is an orthogonal projection. On complex matrix tuples with real Hilbert–Schmidt inner product, define

$$(T_A B)_j = A_j^* B_j + B_j^* A_j.$$

Its real adjoint is $(T_A^* K)_j = 2A_j K_j$. Direct multiplication gives

$$(2.10) \quad \mathcal{C}_E = \frac{1}{4} T_A P T_A^*.$$

This proves self-adjointness, positivity, and (2.8). It also proves the kernel claim. Constant tuples are killed and the displayed formula has zero sum, so no extra projection onto $\mathcal{T}_{d,k}$ is implicit.

In the first line of (2.8), the first term is linear in E and $S_E(K)$ is linear in E . Expanding its squared norm gives (2.9).

If the restriction of $\mathcal{C}_E$ to $\mathcal{T}_{d,k}$ vanishes, subtracting the arithmetic mean from any tuple shows that its form vanishes on all tuples. Take $K_j = I_d$ at one index and zero elsewhere. Its form is $\text{tr}(E_j - E_j^2) = 0$. Since $0 \preceq E_j \preceq I_d$, this forces $E_j^2 = E_j$. Their sum being I_d , they are mutually orthogonal: each positive term in $\sum_{i \neq j} E_j E_i E_j = 0$ must vanish. Conversely, for orthogonal projections the summands $E_j K_j$ have mutually orthogonal ranges, and $\|\sum_j E_j K_j\|_{\text{HS}}^2 = \sum_j \|E_j K_j\|_{\text{HS}}^2$. Their covariance form is zero. $\square$

Lemma 2.3 (Regularized horizontal energy). *For $H \in \mathcal{T}_{d,k}$ and $\tau > 0$,*

$$(2.11) \quad \langle H, (\mathcal{C}_E + \tau \text{Id})^{-1} H \rangle = \min_{\substack{A^*B=0, \\ T_A B + R = H}} \min_{R \in \mathcal{T}_{d,k}} \left(4\|B\|_{\text{HS}}^2 + \tau^{-1}\|R\|_{\text{HS}}^2 \right).$$

If $K = (\mathcal{C}_E + \tau \text{Id})^{-1} H$, a minimizer is

$$(2.12) \quad B_j = \frac{1}{2} A_j (K_j - S_E(K)), \quad R = \tau K.$$

Proof. The constraint $A^*B = 0$ means $PB = B$. Thus $B = PT_A^*K/4$ gives $T_A B = \mathcal{C}_E K$ and is feasible with $R = \tau K$. Its objective is $\langle K, \mathcal{C}_E K \rangle + \tau\|K\|^2 = \langle H, K \rangle$. For any feasible perturbation $(\Delta B, \Delta R)$, its linear objective term is

$$8\langle B, \Delta B \rangle + 2\tau^{-1}\langle R, \Delta R \rangle = 2\langle K, T_A \Delta B + \Delta R \rangle = 0.$$

The remaining terms are nonnegative. This proves minimality, including at singular E ; no inverse of an effect was used. $\square$

2.2. Adaptive differentiation and boundary passage.

Proof of Theorem 2.1. First suppose every effect at the point of differentiation is positive definite. The tangent $H_0 = T_A B$ in (2.12) has zero sum. For small real s , $E_j + s(H_0)_j$ is positive and the tuple remains normalized. If $R_j(s)$ is its positive square root, then

$$U_j = (B_j - R'_j(0))A_j^{-1}$$

is anti-Hermitian. Indeed, multiplication of $U_j + U_j^*$ on the left and right by A_j gives $A_j B_j + B_j^* A_j - A_j R'_j(0) - R'_j(0) A_j = 0$. The factors $\exp(sU_j)R_j(s)$ therefore realize the derivative B_j while preserving the effects exactly.

Use the measurement isometry with an observed label and an inaccessible duplicate label, together with the factor environment. At the point under consideration it obeys

$$W^* \dot{W} = \sum_j A_j^* B_j = 0, \quad \|\dot{W}\|_{\text{op}} \leq \|B\|_{\text{HS}}.$$

Purify any fixed common N -slot tester. In the inner product of terms differentiated at distinct slots, cancel the common later isometries; the later differentiated slot contains $W^* \dot{W}$ or its adjoint and vanishes. Thus the observed output derivative in the direction H_0 has trace norm at most $2\sqrt{N}\|B\|_{\text{HS}}$. This holds for every reference and memory dimension. The environments used for comparison are not returned to the tester.

The output derivative of the same tester is linear in the channel tangent. For the residual tuple R , one call on a joint state ρ has reference blocks $\text{tr}_A[(R_j \otimes I)\rho]$. Duality against Hermitian contractions bounds the trace norm of the j th block by $\|R_j\|_{\text{op}}$. Inserting the residual at each slot and using trace-norm contraction hence costs at most

$$N \sum_j \|R_j\|_{\text{op}} \leq N\sqrt{k}\|R\|_{\text{HS}}.$$

This is a differential decomposition, not a proposed physical mixture of measurements. With $\tau = N^{-1}$, Cauchy-Schwarz and (2.11) now give

$$\begin{aligned} 2\sqrt{N}\|B\|_{\text{HS}} + N\sqrt{k}\|R\|_{\text{HS}} &\leq \sqrt{k+1}\sqrt{4N\|B\|_{\text{HS}}^2 + N^2\|R\|_{\text{HS}}^2} \\ &= \sqrt{k+1} g_{N,E}(H). \end{aligned}$$

Integrate for the fixed tester, then take the supremum. Padding a publicly stopped experiment with discarded dummy calls makes its original stopped output a common postprocessing of an N -slot experiment, so the same bound applies.

For a boundary path, use $E^\epsilon(t) = (1 - \epsilon)E(t) + \epsilon U$, where $U_j = I_d/k$ and $0 < \epsilon < 1$. The covariance is polynomial and positive semidefinite, so its regularized inverse is continuous. Moreover $g_{N,E}(H) \leq N\|H\|_{\text{HS}}$, uniformly on the closed body. Dominated convergence applies to the integrals. The hybrid bound $D_N(E, F) \leq N \sum_j \|E_j - F_j\|_{\text{op}}$ gives convergence of the endpoint distances. This proves (2.6) on the entire body.

For the segment $E(t) = (1 - t)E + tF$, put $M = (E + F)/2$ and $H = F - E$. Concavity and positivity imply, for $0 < t \leq 1/2$,

$$\mathcal{C}_{E(t)} + \tau \text{Id} \succeq 2t(\mathcal{C}_M + \tau \text{Id}).$$

Inverse order bounds $g_{N,E(t)}(H)$ by $(2t)^{-1/2}g_{N,M}(H)$; use $2(1 - t)$ on the second half. The two scalar integrals sum to 2, proving (2.5). Finally, substituting $K = (B, -B)$ into (2.2) gives (2.7). The factor two in the quadratic form comes from the two equal Hilbert-Schmidt contributions. $\square$

2.3. Classical postprocessing.

Proposition 2.4 (Data processing of horizontal energy). *Let $T = (T_{aj})$ be a column-stochastic classical channel and $F_a = \sum_j T_{aj} E_j$. Write $(T^* L)_j = \sum_a T_{aj} L_a$. Then, on all Hermitian tuples,*

$$(2.13) \quad \langle L, \mathcal{C}_F L \rangle \geq \langle T^* L, \mathcal{C}_E(T^* L) \rangle.$$

Consequently the extended quadratic energy

$$\mathcal{I}_E(H) = \sup_K \{2\langle H, K \rangle - \langle K, \mathcal{C}_E K \rangle\}$$

satisfies $\mathcal{I}_F(TH) \leq \mathcal{I}_E(H)$. The value may be infinite at a boundary; when finite it is $\langle H, \mathcal{C}_E^\dagger H \rangle$.

Proof. The two S matrices coincide. The difference of the covariance forms is therefore

$$\sum_j \text{tr} \left[E_j \left(\sum_a T_{aj} L_a^2 - \left(\sum_a T_{aj} L_a \right)^2 \right) \right] \geq 0.$$

The bracket equals the positive sum $\frac{1}{2} \sum_{a,b} T_{aj} T_{bj} (L_a - L_b)^2$. Substitution in the variational definition and enlargement from $K = T^* L$ to all K prove the energy inequality. The pseudoinverse statement follows by diagonalizing the real self-adjoint covariance; a component along its kernel makes the supremum infinite. $\square$

2.4. A uniform noise crossover, sharp in horizon and noise.

Theorem 2.5 (Regularization of arbitrary support patterns). *For $0 \leq \epsilon \leq 1$ put $E^\epsilon = (1-\epsilon)E + \epsilon U$. On $\mathcal{T}_{d,k}$,*

$$(2.14) \quad \mathcal{C}_{E^\epsilon} \succeq (1-\epsilon)\mathcal{C}_E + \frac{\epsilon}{k} \text{Id}.$$

In particular, for arbitrary $E, F \in \mathcal{P}_{d,k}$,

$$(2.15) \quad D_N(E^\epsilon, F^\epsilon) \leq \min \left\{ 2, \frac{2\sqrt{k+1}(1-\epsilon)N}{\sqrt{1+N\epsilon/k}} \|F - E\|_{\text{HS}} \right\}.$$

The dependence on N and ϵ is attained, up to constants depending only on k , by a two-dimensional family. Namely, let $P = |0\rangle\langle 0|$, $P_\theta = |u_\theta\rangle\langle u_\theta|$, $u_\theta = \cos\theta|0\rangle + \sin\theta|1\rangle$, and

$$E = (P, I_2 - P, 0, \dots, 0), \quad F = (P_\theta, I_2 - P_\theta, 0, \dots, 0).$$

For $0 \leq \theta \leq \pi/2$, $0 \leq \epsilon \leq 1/2$, and $z = (1-\epsilon)N \sin\theta / \sqrt{1+N\epsilon}$,

$$(2.16) \quad \frac{\min\{1, z\}}{16384} \leq D_N^{\text{na}}(E^\epsilon, F^\epsilon) \leq D_N(E^\epsilon, F^\epsilon) \leq 4\sqrt{k(k+1)} \min\{1, z\}.$$

Proof. For $K \in \mathcal{T}_{d,k}$, $S_U(K) = 0$ and $\mathcal{C}_U K = K/k$. Concavity proves (2.14). Apply it at the midpoint and use Theorem 2.1 to obtain (2.15).

For the displayed family, keep labels one and two and send every other label to an independent fair bit. This common postprocessing produces the binary effects $B = \beta P + \epsilon I_2/2$ and $B_\theta = \beta P_\theta + \epsilon I_2/2$, where $\beta = 1 - \epsilon$. The equality holds on subnormalized reference states. It therefore preserves the nonadaptive lower witnesses as well as channel postprocessing. The midpoint eigenvalues are $(1 \pm \beta \cos\theta)/2$, and direct substitution in (S4.5) gives

$$Q_N(B, B_\theta)^2 = \frac{4N^2\beta^2 \sin^2\theta}{2 + N(1 - \beta^2 \cos^2\theta)} \geq \frac{4z^2}{2 + z^2}.$$

Here $1 - \beta^2 \leq 2\epsilon$ and $N \geq 1$ were used. It follows that $\min\{1, Q_N(B, B_\theta)\} \geq \min\{1, z\}$. The binary lower theorem in dimension two gives the first inequality of (2.16). For the upper, $\|F - E\|_{\text{HS}} = 2 \sin\theta$ and $(1 + N\epsilon)/(1 + N\epsilon/k) \leq k$. Combine (2.15) with $D_N \leq 2$. $\square$

The family passes from projections and zero effects at $\epsilon = 0$ to strictly positive effects at every $\epsilon > 0$. Thus a bound uniform over the closed body must accommodate both the N scale and the $\sqrt{N}/\epsilon$ scale. This conclusion does not assert a matching entropy law on all multi-outcome boundary strata.

2.5. Exact evaluation without effect square roots.

Proposition 2.6 (Polynomial-size rational certificate). *For Gaussian-rational $E, F \in \mathcal{P}_{d,k}$ and integer $N \geq 1$, $\mathcal{Q}_N(E, F)^2$ is rational and can be evaluated by exact linear algebra on a positive definite real system of dimension $p = (k-1)d^2$. The bit complexity is polynomial in d, k , the total input bit length, and $\log N$. Legality is decidable by exact positive-semidefinite tests. No effect square root, ODE integration, rank decision by tolerance, or enumeration of a dictionary is required.*

Proof. Use the rational Hermitian matrix basis consisting of diagonal units and real and imaginary off-diagonal pairs. For each of the first $k-1$ slots, insert one matrix basis element there and its negative in the last slot. These tuples B_a form a rational basis of $\mathcal{T}_{d,k}$; they need not be orthonormal. Form

$$G_{ab} = \langle B_a, B_b \rangle, \quad L_{ab} = \langle B_a, \mathcal{C}_{(E+F)/2} B_b \rangle, \quad b_a = \langle B_a, F - E \rangle.$$

All entries are rational, G is positive definite, and $L \succeq 0$. Solve $(L + N^{-1}G)x = b$. Then $\mathcal{Q}_N^2 = Nb^\top x$. Omitting the Gram matrix would compute a different regularization. Fraction-free elimination performs polynomially many rational arithmetic operations; clearing input denominators and applying determinant bounds gives polynomial bit length for all minors and for the output. The same reasoning applies to exact symmetric elimination for each effect's positivity. The complexity statement concerns this certificate evaluator, not the exhaustive public code or collective-readout synthesis. $\square$

3. A FINITE PRODUCT ESTIMATE

All classical distances in this paper are unhalved ℓ^1 distances. The following elementary estimate is the Bernoulli-product input to Theorems 5.1 and 9.1. We include its proof so that the support and discrimination theorems can be checked without the binary supplement.

Lemma 3.1 (A uniform finite Bernoulli lower bound). *For $p, q \in [0, 1]$ and every integer $N \geq 1$,*

$$(3.1) \quad \|\text{Ber}(p)^{\otimes N} - \text{Ber}(q)^{\otimes N}\|_1 \geq \frac{1}{128} \min\{1, \sqrt{N}|p - q|\}.$$

In particular, if a differentiable two-sided probability curve satisfies $p'(0) \neq 0$, then, on a sufficiently small two-sided interval, its product experiment separates $p(0)$ from $p(t)$ by at least $c \min\{1, \sqrt{N}|t|\}$ for every N , with $c > 0$ depending on that curve. Necessarily $0 < p(0) < 1$.

Proof. Set $h = |p - q|$. First let $m \geq 1$ and $\sqrt{mh} \leq 1/4$. For $u = m^{-1/2}$ define $z(a) = 1 - a + ae^{iu}$, $0 \leq a \leq 1$. The argument of $z(a)$ has a continuous choice since its real part is positive. Direct differentiation and $0 < u \leq 1$ give

$$|z(a)|^2 \geq 1 - \frac{1}{4m}, \quad \frac{u}{2} \leq \frac{d}{da} \arg z(a) = \frac{\sin u}{|z(a)|^2} \leq \frac{4u}{3}.$$

Bernoulli's inequality gives $|z(a)|^m \geq \sqrt{3/4} > 3/4$. The arguments of $z(p)^m$ and $z(q)^m$ differ by an angle $\vartheta \in [\sqrt{mh}/2, 4\sqrt{mh}/3] \subset [0, 1/3]$. For two complex numbers of moduli at least $3/4$,

$$|z(p)^m - z(q)^m| \geq \frac{3}{2} \sin(\vartheta/2) \geq \frac{3}{2\pi} \vartheta \geq \frac{1}{16} \sqrt{mh}.$$

These complex numbers are the expectations of $\exp(iu \sum_{a=1}^m X_a)$ under the two Bernoulli product laws. This statistic has modulus one, so their ℓ^1 distance is at least its expectation difference.

If $h \leq 1/4$ and $\sqrt{N}h \leq 1/4$, use $m = N$. If $h \leq 1/4$ and $\sqrt{N}h > 1/4$, use $m = \lfloor (16h^2)^{-1} \rfloor \leq N$. Then $1/(4\sqrt{2}) \leq \sqrt{m}h \leq 1/4$, since $\lfloor x \rfloor \geq x/2$ for $x \geq 1$. Discarding the other observations is a common processing and gives a lower bound $1/(64\sqrt{2}) > 1/128$. For $h > 1/4$ the one-copy distance is $2h > 1/2$. The case $h = 0$ is immediate. Finally differentiability gives $|p(t) - p(0)| \geq |p'(0)||t|/2$ near zero. A two-sided probability curve cannot have a nonzero first derivative at either endpoint. $\square$

4. SUPPORT GEOMETRY OF THE COVARIANCE KERNEL

Let $\mathcal{H}_d$ be the real Hilbert space of Hermitian $d \times d$ matrices, with inner product $\text{tr}(AB)$. Throughout this section, $E = (E_1, \dots, E_k)$ is an arbitrary ordered measurement: $E_j \succeq 0$ and $\sum_j E_j = I_d$. Put

$$P_j = \text{supp } E_j, \quad Q_j = I_d - P_j, \quad r_j = \text{rank } E_j.$$

The support of a zero effect is the zero projection. The real subspaces

$$(4.1) \quad \mathcal{W}_E = \sum_{j=1}^k P_j \mathcal{H}_d P_j, \quad \mathcal{A}_E = \mathcal{W}_E^\perp = \{A \in \mathcal{H}_d : P_j A P_j = 0 \text{ for every } j\}$$

will be called the *support span* and its orthogonal complement. The sum in (4.1) is a sum of real linear subspaces, not an assertion that $\mathcal{W}_E$ is an algebra. In particular, for higher-rank effects it need not equal the span of the effects themselves. We use the zero-sum tuple space $\mathcal{T} = \{K \in \mathcal{H}_d^k : \sum_j K_j = 0\}$ with the sum Hilbert–Schmidt inner product, and the covariance of Lemma 2.2.

Theorem 4.1 (Complete support description of the kernel). *For $A \in \mathcal{A}_E$ and $Z_j \in Q_j \mathcal{H}_d Q_j$, set*

$$(4.2) \quad X_j = i[P_j, A] + Z_j, \quad K_j = X_j - \frac{1}{k} \sum_{\ell=1}^k X_\ell.$$

This is a real linear bijection from $\mathcal{A}_E \oplus \bigoplus_j Q_j \mathcal{H}_d Q_j$ onto $\ker(\mathcal{C}_E|_{\mathcal{T}})$. Consequently

$$(4.3) \quad \dim \ker(\mathcal{C}_E|_{\mathcal{T}}) = d^2 - \dim \mathcal{W}_E + \sum_j (d - r_j)^2.$$

In particular the kernel, and not just its dimension, depends only on the ordered support projections.

Proof. Write $S = \sum_j E_j K_j$. The nonnegative variance decomposition in Lemma 2.2 shows that K is in the kernel exactly when

$$(4.4) \quad P_j K_j = P_j S \quad \text{for every } j.$$

Indeed multiplication by $\sqrt{E_j}$ has precisely the same kernel as multiplication by P_j ; no inverse at a singular effect is used. Decompose $S = B + iA$, where A, B are Hermitian. Since $P_j K_j P_j$ and $P_j B P_j$ are Hermitian, compressing (4.4) gives $P_j A P_j = 0$. Furthermore $P_j(K_j - B) = iP_j A$, with the adjoint identity on the other side. These determine the two off-diagonal support blocks and the support block of $K_j - B$; its remaining block is arbitrary. Thus

$$K_j = B + i[P_j, A] + Z_j, \quad Z_j = Q_j Z_j Q_j = Z_j^*.$$

The condition $\sum_j K_j = 0$ forces $B = -k^{-1} \sum_j (i[P_j, A] + Z_j)$, proving necessity.

Conversely, let A, Z_j be as stated. Because $P_j A P_j = 0$, $E_j A P_j = 0$, and hence

$$\sum_j E_j X_j = i \sum_j E_j A = iA.$$

For the mean-subtracted tuple, therefore, $S = iA - \bar{X}$, where $\bar{X} = k^{-1} \sum_j X_j$ is Hermitian. We have $P_j X_j = iP_j A$, which proves (4.4). To check injectivity, suppose that all K_j vanish. Then all X_j equal a common Hermitian matrix D , so the last display gives $D = iA$. A matrix that is both Hermitian and anti-Hermitian is zero. Therefore $A = D = 0$ and all $Z_j = 0$. Finally $\dim(Q_j \mathcal{H}_d Q_j) = (d - r_j)^2$; rank-nullity in (4.1) proves (4.3). $\square$

Proposition 4.2 (Unitary directions and the support span). *Let $G \in \mathcal{H}_d$ and $H_j = i[G, E_j]$. Then*

$$(4.5) \quad H \in \text{Ran}(\mathcal{C}_E|_{\mathcal{T}}) \iff G \in \mathcal{W}_E.$$

More precisely, the pairing of H with the kernel tuple (4.2) is $-2 \text{tr}(GA)$.

Proof. We have $Q_j H_j Q_j = 0$, so the Z_j terms have zero pairing with H ; its zero tuple sum also removes $\bar{X}$. Trace cyclicity gives

$$\sum_j \text{tr}(H_j i[P_j, A]) = \text{tr} \left(A \sum_j i[H_j, P_j] \right).$$

Expanding in the displayed order yields

$$\sum_j i[H_j, P_j] = -2G + \sum_j (E_j G P_j + P_j G E_j).$$

Every summand in the last sum belongs to $P_j \mathcal{H}_d P_j$; it is therefore orthogonal to A . The pairing is $-2 \text{tr}(GA)$ as asserted. A self-adjoint linear operator in finite dimension has range equal to the orthogonal complement of its kernel. Theorem 4.1 now proves (4.5). $\square$

Corollary 4.3 (Regularized tangent scales). *Let Π_0 be the orthogonal projection onto $\ker(\mathcal{C}_E|_{\mathcal{T}})$, and let u_ℓ be an orthonormal basis of its complement with $\mathcal{C}_E u_\ell = \lambda_\ell u_\ell$, $\lambda_\ell > 0$. For every $H \in \mathcal{T}$,*

$$(4.6) \quad g_{N,E}(H)^2 = N^2 \|\Pi_0 H\|_{\text{HS}}^2 + N \sum_\ell \frac{|\langle H, u_\ell \rangle|^2}{\lambda_\ell + N^{-1}}.$$

Thus a nonzero horizontal direction has order $\sqrt{N}$, whereas a direction with nonzero kernel component has order N , at a fixed measurement. For $H = i[G, E]$ the first alternative is exactly $G \in \mathcal{W}_E$ and $H \neq 0$.

Proof. Apply the finite-dimensional spectral theorem to $N(\mathcal{C}_E + N^{-1} \text{Id})^{-1}$ and use Proposition 4.2. $\square$

The corollary concerns a comparison form, not yet an operational lower bound. The distinction matters at the boundary; the next section proves the corresponding finite-pair statement on entire unitary orbits.

Proposition 4.4 (Exact represented-input decision). *On Gaussian-rational effects and a Gaussian-rational Hermitian generator, membership in $\mathcal{W}_E$, stationarity of the orbit, and the kernel dimension (4.3) are decidable by exact rational linear algebra in polynomial bit complexity. The same calculation gives the orthogonal projection of G onto $\mathcal{W}_E$.*

Proof. First use exact positive-semidefinite and normalization tests to validate the input. Select independent columns V_j of E_j by rational elimination. For a nonzero effect its support is

$$P_j = V_j (V_j^* V_j)^{-1} V_j^*;$$

for a zero effect use $P_j = 0$. Compress a rational Hermitian basis by all P_j and select a real-linearly independent subfamily $W_1, \dots, W_q$. The positive Gram system $\sum_b \text{tr}(W_a W_b) c_b = \text{tr}(W_a G)$ gives $\Pi_{\mathcal{W}_E} G = \sum_b c_b W_b$. This tests membership exactly. Commutators test stationarity. Formula (4.3)

then gives the dimension. There are at most kd^2 candidate matrices and d^2 independent matrices. Clearing denominators and applying fraction-free elimination gives polynomial bit bounds by the determinant bounds used in Proposition 2.6. This is a decision about supports and a specified tangent; no optimum over testers or quantum-control synthesis is computed. $\square$

5. FINITE-ANGLE DISTINGUISHABILITY ON UNITARY ORBITS

Fix an ordered measurement E and a Hermitian generator G , and write

$$(5.1) \quad E_j(t) = e^{itG} E_j e^{-itG}, \quad H_j = i[G, E_j].$$

Both E and G are known in this pair-comparison problem. A tester must use the same controls under the two hypotheses E and $E(t)$; its design may depend on this known pair. It has access to the classical device labels and its own reference and memory, never to an unobserved measurement environment.

The Hamiltonian-not-in-Kraus-span criterion and its quantum-error-correction interpretation are established in quantum channel metrology [40, Theorems 1–3]. The support criterion below is its measurement-channel specialization. We prove a finite-angle, finite-use trace-distance statement directly, including an explicit accumulated error estimate. Thus no lower bound is inferred solely from an infinitesimal Fisher-information scaling.

Theorem 5.1 (Finite-use trichotomy on every fixed unitary orbit). *Let $d, N \geq 1$, $k \geq 2$, and let E and G be as in (5.1), with no rank or strict-positivity assumption. The following alternatives are exhaustive.*

- (1) *If $[G, E_j] = 0$ for every j , then $D_N(E, E(t)) = 0$ for all N and all t .*
- (2) *If $H \neq 0$ and $G \in \mathcal{W}_E$, there are constants $c, C, t_0 > 0$, depending on E, G , such that*

$$(5.2) \quad c \min\{1, \sqrt{N}|t|\} \leq D_N(E, E(t)) \leq C \min\{1, \sqrt{N}|t|\}, \quad |t| \leq t_0.$$

The lower bound is attained by a product-input, nonadaptive experiment.

- (3) *If $G \notin \mathcal{W}_E$, there are constants $c, C, t_0 > 0$, depending on E, G , such that*

$$(5.3) \quad c \min\{1, N|t|\} \leq D_N(E, E(t)) \leq C \min\{1, N|t|\}, \quad |t| \leq t_0.$$

The lower bound admits a common adaptive experiment carrying a logical qubit between calls, with a reference of dimension at most $2d$ at an individual call and ideal trusted controls.

The constants are not uniform as the measurement or generator changes. All upper bounds allow arbitrary retained reference and adaptive memory, and bounded public stopping. This theorem classifies a fixed unitary orbit; it does not assert two-sided equivalence of the midpoint covariance modulus on arbitrary pairs in the complete measurement body.

We first record the error-correction statement in the form needed for finite angles. Its exact-correction premise is the usual Knill–Laflamme condition [41]; a proof is included to specify which systems are available to the recovery.

Lemma 5.2 (A corrected logical channel and its quadratic error). *Let $\mathcal{M}$ be a channel on an input space $\mathcal{A}$, and let $J : \mathbb{C}^2 \rightarrow \mathcal{A} \otimes \mathcal{R}$ be an isometry. Suppose a channel $\mathcal{R}$ from the actual output of $\mathcal{M} \otimes \text{Id}_{\mathcal{R}}$ to $\mathbb{C}^2$ satisfies*

$$\mathcal{R} \circ (\mathcal{M} \otimes \text{Id}_{\mathcal{R}}) \circ \mathcal{J} = \text{Id}_2, \quad \mathcal{J}(X) = JXJ^*.$$

For a Hermitian G on $\mathcal{A}$, put $G_L = J^(G \otimes I)J$, $\mathcal{U}_t(X) = e^{-itG_L} X e^{itG_L}$, and*

$$\Phi_t = \mathcal{R} \circ (\mathcal{M} \otimes \text{Id}_{\mathcal{R}}) \circ \text{Ad}(e^{-it(G \otimes I)}) \circ \mathcal{J}.$$

In unhalved diamond norm,

$$(5.4) \quad \|\Phi_t - \mathcal{U}_t\|_\diamond \leq 4t^2 \|G\|_{\text{op}}^2, \quad \|\Phi_t^{\circ m} - \mathcal{U}_t^{\circ m}\|_\diamond \leq 4mt^2 \|G\|_{\text{op}}^2.$$

Proof. Let $\mathcal{T} = \mathcal{R} \circ (\mathcal{M} \otimes \text{Id})$ and choose Kraus operators V_ℓ for $\mathcal{T}$. Since $\mathcal{T} \circ \mathcal{J}$ is the identity channel, whose Choi operator has rank one, $V_\ell J = c_\ell I_2$ with $\sum_\ell |c_\ell|^2 = 1$. Trace preservation on the entire input space gives

$$\sum_\ell c_\ell^* V_\ell = \sum_\ell (V_\ell J)^* V_\ell = J^* \sum_\ell V_\ell^* V_\ell = J^*.$$

It follows by differentiating that $\Phi_0 = \text{Id}_2$ and $\Phi'_0 = -i[G_L, \cdot]$. For any Hermitian B , the generator of $\text{Ad}(e^{-itB})$ has diamond norm at most $2\|B\|_{\text{op}}$. Its second derivative has diamond norm at most $4\|B\|_{\text{op}}^2$, since unitary channels have diamond norm one. Taylor's integral remainder and contractivity under the surrounding channels bound the remainder for Φ_t by $2t^2\|G\|_{\text{op}}^2$. The remainder for $\mathcal{U}_t$ has the same bound because $\|G_L\|_{\text{op}} \leq \|G\|_{\text{op}}$. Their zeroth and first derivatives agree, proving the first inequality. Telescoping the two m -fold compositions of channels proves the second. $\square$

Lemma 5.3 (A reference-preserving code from the support complement). *Suppose $G \notin \mathcal{W}_E$ and put*

$$(5.5) \quad A = G - \Pi_{\mathcal{W}_E} G, \quad s = \text{tr } A_+ = \text{tr } A_- > 0, \quad \rho_\pm = A_\pm / s.$$

There is a code $J : \mathbb{C}^2 \rightarrow \mathbb{C}^d \otimes \mathbb{R}$ with $\dim \mathbb{R} \leq 2d$, corrected exactly by a channel acting only on the classical label and $\mathbb{R}$ after a call to E . The logical generator $G_L = J^(G \otimes I)J$ is diagonal with gap*

$$(5.6) \quad \Delta = \text{tr}(G(\rho_+ - \rho_-)) = \|A\|_{\text{HS}}^2 / s > 0.$$

Proof. The identity belongs to $\mathcal{W}_E$ because $\sum_j E_j = I$; thus A is traceless. Its nonzero positive and negative parts have equal trace. Also

$$P_j \rho_+ P_j = P_j \rho_- P_j \quad \text{for every } j.$$

Purify ρ_+ and ρ_- in $\mathbb{C}^d \otimes \mathbb{C}^d$ and attach orthogonal flags to the second factor. Map the two logical basis vectors to these flagged purifications. They are orthonormal, and the flags make all off-diagonal code matrix elements of $B \otimes I$ vanish. The two diagonal elements are equal for every $B \in P_j M_d(\mathbb{C}) P_j$ by the compression identity above.

Choose measurement Kraus operators $L_{j,a} = |j\rangle\langle a| \sqrt{E_j}$, $1 \leq a \leq d$. Products with different labels are zero; for one label their span is $P_j M_d(\mathbb{C}) P_j$. Hence, with $\mu = (j, a)$,

$$(5.7) \quad J^*(L_\mu^* L_\nu \otimes I)J = \lambda_{\mu\nu} I_2$$

for a positive matrix λ of trace one. Diagonalize λ by a unitary change of Kraus representation. If its positive eigenvalues are λ_b , the corresponding operators $(\tilde{L}_b \otimes I)J/\sqrt{\lambda_b}$ are isometries with mutually orthogonal ranges. A channel projects onto those ranges and applies their inverse isometries; on the orthogonal complement it prepares a fixed logical state. This is completely positive and trace preserving on the entire output space, and it corrects the code exactly. Its input space is just the actual classical-label register tensor $\mathbb{R}$. It does not contain the lost input or a dilation environment. Equivalently, since the actual state is label diagonal, the same channel may first read the label and then act on the retained reference.

The logical generator is diagonal by the flags, with entries $\text{tr}(G\rho_\pm)$. Orthogonality of A to $\Pi_{\mathcal{W}_E} G$ gives $\text{tr}(GA) = \text{tr}(A^2)$, proving (5.6). $\square$

Proof of Theorem 5.1. The first assertion follows directly from (5.1). For the second, Proposition 4.2 gives

$$h^2 = \langle H, \mathcal{C}_E^\dagger H \rangle < \infty.$$

Conjugating all tuple components by e^{itG} intertwines the covariances and their tangent directions, so this value is constant along the orbit. Inverse spectral comparison gives $g_{N,E(t)}(E'(t)) \leq \sqrt{N}h$. The complete-body path theorem therefore yields

$$(5.8) \quad D_N(E, E(t)) \leq \min\{2, \sqrt{k+1} h \sqrt{N}|t|\}.$$

There is an index j with $H_j \neq 0$. Choose a unit vector v with $\langle v, H_j v \rangle \neq 0$. The two Bernoulli parameters obtained by input v and the event “label j ” have gap at least $c_0|t|$ on a sufficiently small two-sided interval. The finite Bernoulli-product bound used in Theorem 9.1 gives $(1/128) \min\{1, c_0 \sqrt{N}|t|\}$. This proves (5.2), including all finite N .

For the third alternative, use Lemma 5.3. The equality $\mathcal{M}_{E(t)} = \mathcal{M}_E \circ \text{Ad}(e^{-itG})$ holds also with a retained reference. Encode, make one device call, recover, and repeat $m \leq N$ times. Under E the logical channel is the identity. Under $E(t)$ it is Φ_t^m from Lemma 5.2. Start in the equal superposition of the two logical basis states. The ideal logical unitary produces unhalved trace distance $2|\sin(mt\Delta/2)|$ from that state. Consequently, for every real t ,

$$(5.9) \quad D_N(E, E(t)) \geq \max_{1 \leq m \leq N} \left(2|\sin(mt\Delta/2)| - 4mt^2 \|G\|_{\text{op}}^2 \right)_+.$$

This bound concerns one specified common tester for each m , not an inaccessible choice of a different recovery under the two hypotheses.

For $t \neq 0$ choose

$$|t| \leq t_0 := \min \left\{ \frac{1}{2\Delta}, \frac{\Delta}{4\pi \|G\|_{\text{op}}^2} \right\}, \quad m = \min \left\{ N, \left\lfloor \frac{1}{|t|\Delta} \right\rfloor \right\}.$$

Then $x = m|t|\Delta \leq 1$ and $x \geq \frac{1}{2} \min\{1, N|t|\Delta\}$. Since $2\sin(x/2) \geq 2x/\pi$ and the error in (5.9) is at most x/π , we obtain

$$(5.10) \quad D_N(E, E(t)) \geq \frac{1}{2\pi} \min\{1, N|t|\Delta\}.$$

The elementary hybrid estimate for the input unitaries gives $D_N(E, E(t)) \leq \min\{2, 2N|t|\|G\|_{\text{op}}\}$, which proves (5.3). The zero-angle case is immediate. The protocol uses deterministic $m \leq N$ calls, hence belongs to the class allowing bounded public stopping. The upper estimates hold for that entire class. Finally if the first alternative held with $G \notin \mathcal{W}_E$, then $H = 0$ would contradict Proposition 4.2; the alternatives are disjoint. $\square$

Corollary 5.4 (Full-rank, rank-one and projective measurements). *If one effect is positive definite, every nonconstant unitary orbit has the square-root law (5.2). If all nonzero effects have rank one, every nonconstant orbit has that law precisely when the measurement is informationally complete, meaning that its effects span $\mathcal{H}_d$. If such a rank-one measurement is not informationally complete, some generator has the linear law (5.3). For a projection-valued measurement every orbit is either constant or has the linear law.*

Proof. A full-rank support contributes $\mathcal{H}_d$ to $\mathcal{W}_E$. For rank-one supports, $P_j \mathcal{H}_d P_j = \mathbb{R}E_j$, so that $\mathcal{W}_E$ is exactly the effect span. If this span is proper, a nonzero element of its orthogonal complement is a generator outside $\mathcal{W}_E$ and cannot commute with all effects by Proposition 4.2. For a projection-valued measurement $\mathcal{W}_E$ is the block-diagonal Hermitian space of the orthogonal supports, which is exactly the space of generators commuting with every effect. Apply Theorem 5.1. $\square$

Proposition 5.5 (Support spans under output processing). *Let $T_{aj} \geq 0$ satisfy $\sum_a T_{aj} = 1$, and put $F_a = \sum_j T_{aj} E_j$. Then $\mathcal{W}_E \subseteq \mathcal{W}_F$. Thus classical processing of the ordered output cannot turn a nonconstant square-root orbit into a linear orbit. A reversible splitting of labels preserves the support span and the classification.*

Proof. For positive matrices, the kernel of a positive weighted sum is the intersection of the kernels of its positively weighted summands. Thus the support of F_a contains P_j whenever $T_{aj} > 0$. Every column has such a row, giving $P_j \mathcal{H}_d P_j \subseteq \mathcal{W}_F$. This proves the inclusion. The orbit commutes with classical processing, so Theorem 5.1 applies with the same generator. A reverse stochastic map gives the opposite inclusion for a reversible splitting. $\square$

Remark 5.6 (What the lower construction does not assert). The code and recovery depend on the known pair through E, G . They do not constitute a common estimator of an unknown measurement. Their existence is proved in finite-dimensional quantum mechanics; polynomial gate synthesis, physical execution, and parameter-uniform bounds for approximate controls are not inferred. Formula (5.9) quantifies the finite-angle channel remainder under ideal controls. It is not a claim of perfect finite-use discrimination. The full-body entropy and growing- k learning questions are separate from this orbit classification.

6. FINITE-USE CLASSIFICATION OF SMOOTH BOUNDARY CURVES

The support criterion extends beyond unitary conjugation. Here $E(t) = (E_1(t), \dots, E_k(t))$, $|t| < a$, is a fixed two-sided C^2 curve of ordered measurements, with $E = E(0)$ and $H = E'(0) \neq 0$. The curve may change ranks at zero and need not have a differentiable exact Kraus representation there. All constants below belong to this fixed curve, not uniformly to the measurement body.

Lemma 6.1 (The support generator of a two-sided tangent). *Let $P_j = \text{supp } E_j$ and $Q_j = I - P_j$. Then $Q_j H_j Q_j = 0$. Conversely every Hermitian zero-sum tuple with these vanishing blocks is the derivative of a real-analytic two-sided measurement curve. Define the Hermitian matrix*

$$(6.1) \quad \Gamma_E(H) = \frac{i}{2} \sum_j [P_j, H_j].$$

On the real zero-sum tuple space,

$$(6.2) \quad H \in \text{Ran } \mathcal{C}_E \iff \Gamma_E(H) \in \mathcal{W}_E.$$

For a unitary tangent $H_j = i[G, E_j]$, $\Gamma_E(H) - G \in \mathcal{W}_E$. In particular every stationary unitary generator belongs to $\mathcal{W}_E$.

Proof. For $v \in \text{Ran } Q_j$, the nonnegative function $\langle v, E_j(t)v \rangle$ has a two-sided minimum at zero; its derivative vanishes. Polarization on that subspace gives $Q_j H_j Q_j = 0$. The sum of the H_j vanishes by normalization. Pairing with the complete kernel tuple in Theorem 4.1, both the mean and the missing-support blocks disappear, leaving

$$\langle H, K(A, Z) \rangle = \text{tr} \left(A \sum_j i[H_j, P_j] \right) = -2 \text{tr}(A \Gamma_E(H)).$$

Finite-dimensional self-adjointness proves (6.2). For a unitary tangent, expansion gives

$$\Gamma_E(H) = G - \frac{1}{2} \sum_j (E_j G P_j + P_j G E_j),$$

and the last sum belongs to $\mathcal{W}_E$. If the tangent is zero, this identity also proves the final assertion. For the converse realizability statement, the normalized-factor construction in the proof of Lemma 6.4 is real analytic near zero and has precisely the prescribed value and derivative. $\square$

Theorem 6.2 (Two-sided smooth-curve classification). *For the fixed C^2 curve above there are $c, C, t_0 > 0$ such that, for all integers $N \geq 1$ and $|t| \leq t_0$,*

$$(6.3) \quad c \min\{1, N^\alpha |t|\} \leq D_N(E, E(t)) \leq C \min\{1, N^\alpha |t|\}, \quad \alpha = \begin{cases} 1/2, & \Gamma_E(H) \in \mathcal{W}_E, \\ 1, & \Gamma_E(H) \notin \mathcal{W}_E. \end{cases}$$

The first lower bound uses repeated product inputs. The second has a known-pair adaptive realization with ideal encoding and recovery, a logical qubit, and a per-call external reference of dimension at most $2d$. The same controls are used under both hypotheses. Constants may degenerate with the tangent, nonzero support eigenvalues, curvature, or support-complement gap.

The exponents and the error-correction mechanism have the established channel-metrology antecedents [40, 41]. The content here is a direct finite-pair theorem from C^2 effect coordinates: the support formula decides its branch even at second-order rank openings, without assuming a smooth exact Kraus family or inferring a finite-angle lower from Fisher information. It is a theorem for each fixed nonzero first-order curve, not a uniform arbitrary-pair midpoint equivalence.

Lemma 6.3 (A horizontal curve with prescribed first derivative). *If $H \in \text{Ran } \mathcal{C}_E$, there are factors $A_j = \sqrt{E_j}$ and B_j satisfying*

$$A^*A = I, \quad A^*B = 0, \quad A_j^*B_j + B_j^*A_j = H_j,$$

where A, B denote the vertically stacked matrices. Put $b = \|B\|_{\text{op}}$ and $S = B^*B$. The factors

$$(6.4) \quad A(t) = (A + tB)(I + t^2S)^{-1/2}$$

define a normalized measurement $\tilde{E}_j(t) = A_j(t)^*A_j(t)$ with the same zeroth and first derivatives as $E(t)$, and

$$A(t)^*A'(t) = 0, \quad \|A'(t)\|_{\text{op}} \leq b, \quad \|A''(t)\|_{\text{op}} \leq 7b^2 \quad (|t|b \leq 1).$$

If $L = \sup_{|s| \leq a_0} \sum_j \|E_j''(s)\|_{\text{op}}$ on a closed subinterval of the given curve, then, for $|t| \leq a_0$ and $|t|b \leq 1$,

$$(6.5) \quad D_N(E, E(t)) \leq \min \left\{ 2, 2b\sqrt{N}|t| + \frac{1}{2}(L + 16b^2)Nt^2 \right\}.$$

Proof. The factorization $\mathcal{C}_E = T_A(I - AA^*)T_A^*/4$ in Lemma 2.2 gives a horizontal solution: for $K = \mathcal{C}_E^\dagger H$ take $B = (I - AA^*)T_A^*K/4$. Thus $T_AB = H$ and $A^*B = 0$. In fact $4\|B\|_{\text{HS}}^2 = \langle H, \mathcal{C}_E^\dagger H \rangle$. Formula (6.4) is normalized because $(A + tB)^*(A + tB) = I + t^2S$. Its derivative is

$$A'(t) = (B - tAS)(I + t^2S)^{-3/2}, \quad A'(t)^*A'(t) = S(I + t^2S)^{-2}.$$

These identities prove horizontality and the first derivative bound. Differentiation once more gives, in the displayed order,

$$A''(t) = -AS(I + t^2S)^{-3/2} - 3t(B - tAS)S(I + t^2S)^{-5/2}.$$

Since $\|A\| = 1$, $\|S\| = b^2$, and $|t|b \leq 1$, its norm is at most $b^2 + 3|t|(b + |t|b^2)b^2 \leq 7b^2$.

Duplicate each output label into an inaccessible environment to form a Stinespring isometry $W(t)$ from $A(t)$. It has exactly the same operator derivative bounds and $W(t)^*W'(t) = 0$. For one fixed purified adaptive tester, differentiating its final vector gives a sum over slots. Distinct terms are orthogonal: after cancelling later common isometries, the later differentiated slot contains W^*W' or its adjoint. Each diagonal term has squared norm at most b^2 . The final vector derivative is at most $\sqrt{N}b$, and the density derivative at most $2\sqrt{N}b$. Integration followed by common partial trace proves $D_N(E, \tilde{E}(t)) \leq 2b\sqrt{N}|t|$. Padding stopped branches by discarded fixed-input calls includes bounded public stopping; the original stopped output is a common postprocessing of the padded experiment.

For a Hermitian tuple R , the measurement derivative has diamond norm at most $\sum_j \|R_j\|_{\text{op}}$. Indeed each reference output block is the partial trace against R_j ; its trace norm on a state is at most $\|R_j\|_{\text{op}}$ by duality. The Hermiticity-preserving diamond norm may be optimized over states with a reference. The Stinespring second derivative bound similarly gives $\|\mathcal{M}_{\tilde{E}(t)}''\|_{\diamond} \leq 2\|W''(t)\| + 2\|W'(t)\|^2 \leq 16b^2$. Taylor's integral formula, with common value and derivative at

zero, therefore bounds the one-call channel difference by $(L + 16b^2)t^2/2$. Hybrid telescoping over N calls proves (6.5). $\square$

Lemma 6.4 (A logical derivative for a general measurement curve). *Suppose $\Gamma = \Gamma_E(H) \notin \mathcal{W}_E$. Use the support-complement code of Lemma 5.3 with Γ in place of G , and let $\mathcal{R}$ be its fixed CPTP recovery. Put*

$$\Phi_t = \mathcal{R} \circ (\mathcal{M}_{E(t)} \otimes \text{Id}) \circ \mathcal{J}, \quad \Gamma_L = J^*(\Gamma \otimes I)J.$$

Then $\Phi_0 = \text{Id}_2$, $\Phi'_0 = -i[\Gamma_L, \cdot]$, and, for $|t| \leq a_0$,

$$(6.6) \quad \|\Phi_t - \text{Ad}(e^{-it\Gamma_L})\|_{\diamond} \leq \kappa t^2, \quad \kappa = \frac{1}{2}L + 2\|\Gamma\|_{\text{op}}^2.$$

Its logical gap is $\Delta = \|Z\|_{\text{HS}}^2 / \text{tr } Z_+ > 0$, where $Z = \Gamma - \Pi_{\mathcal{W}_E}\Gamma$.

Proof. To compute the derivative, not to factor the unknown curve itself, put

$$A_j = \sqrt{E_j}, \quad B_j = (\sqrt{E_j})^+ H_j (I - P_j/2).$$

The inverse here is only on the fixed support; it is zero on its complement. Since $Q_j H_j Q_j = 0$, $A_j^* B_j + B_j^* A_j = H_j$ and

$$\sum_j A_j^* B_j = \frac{1}{2} \sum_j [P_j, H_j] = -i\Gamma.$$

In particular this sum is anti-Hermitian. The normalized factors $(A + tB)(I + t^2 B^* B)^{-1/2}$ have derivative B and induce the same channel value and derivative as $E(t)$, irrespective of its second-order rank changes.

Write $L_\mu = |j\rangle\langle a| A_j$ and $\dot{L}_\mu = |j\rangle\langle a| B_j$ for $\mu = (j, a)$, with identity on the reference understood. If R_ℓ are Kraus operators of the completed recovery, exact correction gives $R_\ell L_\mu J = c_{\ell\mu} I_2$. The left first-order term of the logical channel is therefore

$$\sum_{\ell, \mu} \bar{c}_{\ell\mu} R_\ell \dot{L}_\mu J = J^* \sum_{\mu} L_\mu^* \dot{L}_\mu J = -i\Gamma_L,$$

where completeness of the recovery is used in the first equality. The adjoint term proves the commutator derivative. The actual logical second derivative is at most L in diamond norm, and that of the logical unitary is at most $4\|\Gamma\|_{\text{op}}^2$. Taylor's integral remainder proves (6.6). The support-code lemma gives the gap and the available systems. $\square$

Proof of Theorem 6.2. If H is in the covariance range, let $x = \sqrt{N}|t|$. For $x \leq 1$, (6.5) is at most $[2b + (L + 16b^2)/2]x$; for $x \geq 1$ use the bound two. For the lower, choose j, v with $\langle v, H_j v \rangle \neq 0$. The fixed input v and the event “label j ” have a first-order probability gap, and Lemma 3.1 proves the result.

If H is outside the range, apply Lemma 6.4. Starting from the equal logical superposition and telescoping m corrected calls gives, for $|t| \leq a_0$,

$$(6.7) \quad D_N(E, E(t)) \geq \max_{1 \leq m \leq N} (2|\sin(mt\Delta/2)| - m\kappa t^2)_+.$$

Choose

$$t_0 \leq \min\{a_0, (2\Delta)^{-1}, \Delta/(2\pi\kappa)\}, \quad m = \min\{N, \lfloor (|t|\Delta)^{-1} \rfloor\}.$$

For $0 < |t| \leq t_0$ the floor is at least two, and $x = m|t|\Delta$ satisfies $\frac{1}{2} \min\{1, N|t|\Delta\} \leq x \leq 1$. The sine term is at least $2x/\pi$ and the error at most $x/(2\pi)$. This gives a positive constant times $\min\{1, N|t|\}$. For the upper, the finite constant $L_1 = \sup_{|s| \leq a_0} \sum_j \|E'_j(s)\|_{\text{op}}$ and hybrid telescoping give $D_N \leq \min\{2, NL_1|t|\}$. The point $t = 0$ is immediate. $\square$

Corollary 6.5 (Rank opening without a unitary orbit). *Let $P = |0\rangle\langle 0|$ on $\mathbb{C}^2$, $G = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$, $U_t = e^{itG}$, and for $|t| < 1/2$ set*

$$E_1(t) = (1 - t^2)U_t P U_t^* + \frac{1}{2}t^2 I_2, \quad E_2(t) = I_2 - E_1(t).$$

Both effects are full rank for $t \neq 0$ and projective at zero. Nevertheless $D_N(E(0), E(t)) \asymp \min\{1, N|t|\}$ on a fixed small interval, uniformly over $N \geq 1$.

Proof. At zero the derivative is $i[G, (P, I - P)] \neq 0$ and the support span is diagonal, while G is not. Lemma 6.1 and Theorem 6.2 apply. The eigenvalues of the first effect are $t^2/2$ and $1 - t^2/2$, proving the rank assertion. $\square$

A vanishing first derivative is intentionally excluded from the classification: it does not imply stationarity of a general curve. For example the scalar curve $(t^2, 1 - t^2)$ has distance $2[1 - (1 - t^2)^N]$ from $(0, 1)$, of order $\min\{1, Nt^2\}$. Likewise a one-sided rank opening can have a nonzero missing-support first-order block and is not covered by the two-sided tangent lemma.

7. CERTIFIED ERRORS IN THE DISCRIMINATION CONTROLS

The preceding lower bounds first concern ideal known-pair controls. Their stability has a direct finite formulation. An error bound on a control channel is a bound in unhalved diamond norm, valid on every input including its memory reference. It is not a bound measured only on one calibration state.

Theorem 7.1 (A finite control-error ledger). *Use the code and recovery from Lemma 6.4, with gap Δ and remainder constant κ . Suppose an implemented initial logical state differs from the ideal one in trace norm by at most e_0 . At cycle r , suppose its implemented logical channel differs from the corresponding ideal corrected channel by at most ν_r under each hypothesis. Suppose the implemented final readout differs from the ideal Helstrom readout by at most e_f in diamond norm. For every integer $m \leq N$ and $|t| \leq a_0$, the actual classical output laws obey*

$$(7.1) \quad \|\tilde{p}_{0,m} - \tilde{p}_{t,m}\|_1 \geq \left(2|\sin(mt\Delta/2)| - m\kappa t^2 - 2e_0 - 2e_f - 2 \sum_{r=1}^m \nu_r \right)_+.$$

Here one may choose the Helstrom readout for the ideal unrotated and logical-unitary states; it depends on the known pair and on m . The cycle error may be bounded by the sum of the encoding and recovery errors, because the unknown measurement itself is a channel.

In particular, put $z = \min\{1, N|t|\Delta\}$ and choose m, t_0 as in the proof of Theorem 6.2. For $0 < |t| \leq t_0$, the sufficient conditions

$$(7.2) \quad e_0 + e_f \leq z/(16\pi), \quad \nu_r \leq |t|\Delta/(8\pi) \quad (1 \leq r \leq m)$$

imply $\|\tilde{p}_{0,m} - \tilde{p}_{t,m}\|_1 \geq z/(2\pi)$. All errors are deterministic uniform channel bounds; no independence of control errors is required.

Proof. Choose the ideal two-outcome Helstrom readout for the equal logical superposition and its image under $e^{-imt\Gamma_L}$. Its classical ℓ^1 separation is $2|\sin(mt\Delta/2)|$. Replacing the logical unitary by the actual ideal corrected channel loses at most $m\kappa t^2$, by channel telescoping and Lemma 6.4. Under either hypothesis, telescoping the implemented preparation, cycles and readout changes its output law in ℓ^1 by at most $e_0 + e_f + \sum_r \nu_r$. The triangle inequality proves (7.1). Replacing encoding and recovery around one unchanged unknown channel costs no more than the sum of their diamond errors, which proves the stated decomposition of ν_r .

For the indicated choice set $x = m|t|\Delta$, so $z/2 \leq x \leq 1$. The ideal sine term minus curvature is at least $3x/(2\pi)$. Conditions (7.2) give $e_0 + e_f \leq x/(8\pi)$ and $\sum_r \nu_r \leq x/(8\pi)$. The total loss is therefore at most $x/(2\pi)$, leaving $x/\pi \geq z/(2\pi)$. $\square$

For a unitary orbit, Lemma 5.2 gives the same result with $\kappa = 4\|G\|_{\text{op}}^2$ and the original support-code gap. An error allowance fixed independently of the angle need not meet (7.2) as $t \rightarrow 0$. The theorem does not certify a particular piece of hardware, synthesize a gate circuit, or allow target-dependent advice in common learning. It specifies exactly which certified approximation errors suffice to preserve the finite-use lower bound.

Proposition 7.2 (Exact first-order decisions). *For Gaussian-rational E and a Gaussian-rational Hermitian zero-sum H , two-sided first-order realizability and the branch of Theorem 6.2 for $H \neq 0$ are decidable in polynomial represented-input bit complexity. The decision gives $\Gamma_E(H)$, its projection on $\mathcal{W}_E$, and its orthogonal residual. A zero tangent is reported as a higher-order question, not as stationarity of an unspecified curve.*

Proof. The exact support projectors in Proposition 4.4 are rational. Test $Q_j H_j Q_j = 0$ for every j and compute (6.1). Necessity and sufficiency of the test follow from Lemma 6.1 and the normalized-factor construction in Lemma 6.4: for any such tangent it gives a real-analytic measurement curve with that derivative. Real coordinate rank selection and the Hilbert–Schmidt Gram system then project $\Gamma_E(H)$ exactly. Rational matrix arithmetic and fraction-free elimination have the bit bounds already proved in Proposition 4.4. Only these finite operations are performed. A numerical curvature bound for an unspecified curve, an optimal tester, and a compiled recovery are not outputs of this decision. $\square$

8. REGULARIZATION TRANSPORTED BY THE OUTPUT CHANNEL

The fixed Euclidean ridge in Theorem 2.1 is tied to the displayed outcome coordinates. Its contraction under arbitrary stochastic relabeling does not follow from the unregularized data-processing theorem. A ridge obtained from a public probability vector has a different, covariant property.

For a probability vector $w = (w_j)$, put $W = (w_j I_d)_j$ and $\mathcal{B}_w = \mathcal{C}_W$. On the full real Hermitian tuple space this operator is

$$(8.1) \quad (\mathcal{B}_w K)_j = w_j \left(K_j - \sum_{\ell} w_{\ell} K_{\ell} \right).$$

For a zero-sum direction H and $\tau > 0$ define the extended energy

$$(8.2) \quad \mathcal{E}_{E,w,\tau}(H) = \sup_{K \in \mathcal{H}_d^k} \{2\langle H, K \rangle - \langle K, (\mathcal{C}_E + \tau \mathcal{B}_w) K \rangle\}.$$

It equals the pseudoinverse quadratic form on the range and is $+\infty$ off the range. Constant tuples have zero energy and zero pairing with H . If every $w_j > 0$, the supremum is finite and may be evaluated by restricting to $\mathcal{T}$.

Theorem 8.1 (Processing-covariant regularized energy). *For every column-stochastic output map T , every $\tau > 0$, and every zero-sum Hermitian tuple H ,*

$$(8.3) \quad \mathcal{E}_{TE, Tw, \tau}(TH) \leq \mathcal{E}_{E, w, \tau}(H).$$

Zero entries of Tw are allowed, with the extended-energy convention in (8.2). If a stochastic map S satisfies $ST = \text{Id}$, equality holds in (8.3). In particular, reversible outcome splitting leaves this energy unchanged when the public weights are split by the same rule.

Proof. The matrix Jensen identity in Proposition 2.4 is an inequality on the full tuple space, not only on its zero-sum subspace. It gives

$$\langle L, \mathcal{C}_{TE} L \rangle \geq \langle T^* L, \mathcal{C}_E T^* L \rangle.$$

Apply the same inequality to the scalar measurement W to get $\langle L, \mathcal{B}_{Tw} L \rangle \geq \langle T^* L, \mathcal{B}_w T^* L \rangle$. Also $\langle TH, L \rangle = \langle H, T^* L \rangle$. Thus each expression in the supremum defining the left side of (8.3) is bounded by a permissible expression on the right. Taking suprema proves the claim, including infinite values. No assumption that $T^* L$ has zero sum is needed. If $ST = \text{Id}$, application of the same inequality to S gives the reverse bound. $\square$

Proposition 8.2 (Relation to the operational upper certificate). *If $w_j > 0$ and $w_{\max} = \max_j w_j$, then for every pair of legal measurements, with midpoint M and difference H ,*

$$(8.4) \quad D_N(E, F) \leq \min \left\{ 2, 2\sqrt{k+1} \sqrt{N \mathcal{E}_{M,w,1/(Nw_{\max})}(H)} \right\}.$$

For $w_j = 1/k$, one has $\mathcal{B}_w|_{\mathcal{T}} = \text{Id}/k$, and $\mathcal{E}_{M,w,k/N}(H) = \mathcal{Q}_N(E, F)^2/N$ exactly.

Proof. The variance expression gives $0 \preceq \mathcal{B}_w \preceq w_{\max} \text{Id}$. On $\mathcal{T}$, all coordinates of w positive make $\mathcal{B}_w$ positive definite. Inverse order therefore gives

$$\langle H, (\mathcal{C}_M + N^{-1} \text{Id})^{-1} H \rangle \leq \mathcal{E}_{M,w,1/(Nw_{\max})}(H).$$

Use Theorem 2.1. The uniform formula follows by substitution into (8.1). $\square$

Theorem 8.1 compares the same τ and *transported* weights. Replacing Tw by a new uniform vector, or changing τ to the new alphabet size divided by N , is a different operation. Consequently this result does not retroactively assert arbitrary-processing monotonicity of the original fixed ridge. Exact evaluation with positive rational weights uses the tangent Gram system of Proposition 2.6, with the ridge stiffness $\tau \mathcal{B}_w$ in place of G/N .

9. FINITE-OUTCOME GEOMETRY AND EXACT DESCRIPTIONS

The interior geometry does not require two outcomes. We now allow any fixed finite alphabet, while retaining the input-consuming, ordered, classical-output interface. The effects below need not commute. The binary results on the complete effect body remain separate: no positive lower spectral bound is imposed in those results.

For integers $d \geq 1$ and $k \geq 2$, put

$$(9.1) \quad \mathfrak{P}_{d,k} = \left\{ \mathbf{E} = (E_1, \dots, E_k) : E_j = E_j^*, \quad \sum_{j=1}^k E_j = I_d, \quad \frac{I_d}{2k} \preceq E_j \preceq \frac{3I_d}{2k} \right\}.$$

An element specifies the channel $\mathcal{M}_{\mathbf{E}}(\rho) = \sum_j \text{tr}(E_j \rho) |j\rangle\langle j|$. It returns no residual quantum system. Write D_N for its N -use distance, with precisely the reference, feedback, bounded public stopping, and unhalved trace-norm conventions defining d_N . The superscript na denotes nonadaptive experiments, not a restriction to unentangled inputs. Set

$$r(\mathbf{E}, \mathbf{F}) = \max_j \|E_j - F_j\|_{\text{op}}, \quad p = (k-1)d^2.$$

The affine space $\sum_j E_j = I_d$ has real dimension p . In its translated vector space $\sum_j H_j = 0$, the set $\mathfrak{P}_{d,k}$ is exactly the closed r -ball of radius $R = 1/(2k)$ about $\mathbf{U} = (I_d/k, \dots, I_d/k)$.

Theorem 9.1 (A dimension-independent interior comparison). *For $d, N \geq 1$, $k \geq 2$, and $\mathbf{E}, \mathbf{F} \in \mathfrak{P}_{d,k}$,*

$$(9.2) \quad \frac{1}{128} \min\{1, \sqrt{N} r(\mathbf{E}, \mathbf{F})\} \leq D_N^{\text{na}}(\mathbf{E}, \mathbf{F}) \leq D_N(\mathbf{E}, \mathbf{F}) \leq \min\{2, 2k\sqrt{N} r(\mathbf{E}, \mathbf{F})\}.$$

The lower bound holds for arbitrary legal ordered k -outcome measurements, without the spectral bounds in (9.1). The upper constants are independent of d and N ; dependence on k is displayed and is not asserted sharp.

Proof. Let $H_j = F_j - E_j$ and $A_j(t) = E_j + tH_j$ for $0 \leq t \leq 1$. Each $A_j(t)$ is positive definite. Solve the matrix differential equation

$$(9.3) \quad \dot{X}_j(t) = \frac{1}{2}X_j(t)A_j(t)^{-1}H_j, \quad X_j(0) = \sqrt{E_j}.$$

Its coefficients are continuous. Differentiating $X_j^*X_j$ shows that it solves the linear equation

$$\dot{B}_j = \frac{1}{2}H_jA_j^{-1}B_j + \frac{1}{2}B_jA_j^{-1}H_j, \quad B_j(0) = E_j.$$

The function A_j solves this same equation, so uniqueness gives $X_j^*X_j = A_j$. Consequently

$$(9.4) \quad X_j^*\dot{X}_j = \frac{1}{2}H_j, \quad \dot{X}_j^*\dot{X}_j = \frac{1}{4}H_jA_j^{-1}H_j.$$

This argument neither diagonalizes the tuple nor assumes that A_j and H_j commute.

Use the Stinespring isometry

$$W_t\psi = \sum_{j=1}^k |j\rangle_Y |j\rangle_Z X_j(t)\psi.$$

Tracing out Z and the last factor gives $\mathcal{M}_{A(t)}$. The environment in this dilation is used only in the proof and is not supplied by the device. Since $\sum_j H_j = 0$, (9.4) yields

$$(9.5) \quad W_t^*W_t = I_d, \quad W_t^*\dot{W}_t = 0, \quad \dot{W}_t^*\dot{W}_t = \frac{1}{4}\sum_j H_jA_j(t)^{-1}H_j.$$

For completeness, purify a common adaptive tester, retaining a fresh inaccessible environment for each call. Its controls are isometries independent of t . Differentiate the final purified state. It is a sum of at most N terms, with $\dot{W}_t$ replacing one occurrence of W_t in each term. In the inner product of two distinct terms, cancel the later common isometries. At the later differentiated slot the remaining contraction is $W_t^*\dot{W}_t$ or its adjoint, hence zero. Each diagonal term has norm at most $\|\dot{W}_t\|_{\text{op}}$. The derivative of the purified state therefore has norm at most $\sqrt{N}\|\dot{W}_t\|_{\text{op}}$. The derivative of its rank-one density operator has trace norm at most twice this number. Partial trace and all final processing contract trace norm. Bounded stopping is covered by a coherent stopping flag and dummy calls on a fixed discarded input after stopping; the resulting final state of the original tester is a common postprocessing of this padded experiment.

Integrating the derivative bound gives the more informative inequality

$$(9.6) \quad \begin{aligned} D_N(\mathbf{E}, \mathbf{F}) &\leq \sqrt{N} \int_0^1 \left\| \sum_j H_j A_j(t)^{-1} H_j \right\|_{\text{op}}^{1/2} dt \\ &\leq \sqrt{2kN} \left\| \sum_j H_j^2 \right\|_{\text{op}}^{1/2} \leq \sqrt{2k}\sqrt{N} \max_j \|H_j\|_{\text{op}}. \end{aligned}$$

We used $A_j(t)^{-1} \preceq 2kI_d$ and the order-preserving map $B \mapsto H_j B H_j$; commutation is unnecessary. The weaker constant $2k$ in (9.2) is convenient below. The trace distance is always at most two.

For the lower bound, choose a label j and a unit eigenvector of $F_j - E_j$ attaining its operator norm. Send that same vector in every call and replace the observed label by the indicator of j . The two records are products of Bernoulli laws with parameters $s, t \in [0, 1]$ and $|s - t| = \|F_j - E_j\|_{\text{op}}$. Since $\max\{s(1-s), t(1-t)\} + N^{-1} \leq 5/4$, the quantity in (S16.2) is at least $\frac{1}{2}\sqrt{N}|s - t|$. Lemma S16.2 gives the lower constant $1/128$. Maximize over j . No interior assumption was used in this part. $\square$

9.1. The affine norm ball and its entropy. Let $\mathcal{C}_{N,k}(\delta)$ be the least number of radius- δ D_N -balls covering $\mathfrak{P}_{d,k}$. Their centres may be arbitrary legal memoryless ordered k -outcome measurements.

Theorem 9.2 (Dimension-uniform finite-outcome entropy). *For $d, N \geq 1$, $k \geq 2$, and $0 < \delta < 1/128$,*

$$(9.7) \quad \max \left\{ 1, \left(\frac{\sqrt{N}}{256k\delta} \right)^p \right\} \leq \mathcal{C}_{N,k}(\delta) \leq \left(\frac{3\sqrt{N}}{\delta} \right)^p, \quad p = (k-1)d^2.$$

In particular, uniformly in d, N, δ for each fixed k ,

$$(9.8) \quad \log_2 \mathcal{C}_{N,k}(\delta) = \frac{p}{2} \log_2 N + p \log_2(1/\delta) + O(p \log(k+1)).$$

Proof. If $D_N(\mathbf{E}, \mathbf{F}) \leq \delta < 1/128$, the lower bound of Theorem 9.1 implies $r(\mathbf{E}, \mathbf{F}) \leq 128\delta/\sqrt{N}$. Lebesgue volume in the p -dimensional translated affine space gives the lower covering bound, because the target is its r -ball of radius $1/(2k)$. This argument also applies when the centre lies outside $\mathfrak{P}_{d,k}$.

Take a maximal ϵ -separated subset of that ball, where $\epsilon = \delta/(2k\sqrt{N})$. It is an r -net. The open r -balls of radius $\epsilon/2$ about its points are disjoint and lie in the ball of radius $R + \epsilon/2$. Their number is at most $(1 + 2R/\epsilon)^p = (1 + 2\sqrt{N}/\delta)^p$. By the upper comparison it is a D_N -cover at radius δ . This proves (9.7); taking logarithms gives (9.8), including the range where the displayed power in the lower bound is less than one. $\square$

9.2. A finite grid with exact normalization. An independent rounding of all k effects need not preserve their sum. The following grid preserves both normalization and the two spectral inequalities simultaneously.

Lemma 9.3 (Rational nets of the balanced body). *Let $0 < t \leq 1/(8k)$ be rational and choose*

$$(9.9) \quad K = \left\lceil \max \left\{ \frac{4(k-1)d}{t}, 8k(k-1)d \right\} \right\rceil.$$

Enumerate the first $k-1$ Hermitian matrices using diagonal coordinates $0, 1/K, \dots, 1$ and real and imaginary coordinates above the diagonal in $K^{-1}\mathbb{Z} \cap [-1, 1]$. Set the last matrix equal to I_d minus their sum and retain exactly the legal tuples in $\mathfrak{P}_{d,k}$. This finite ordered grid, denoted $\mathcal{G}(t)$, is an r -net of radius t . All membership and pairwise norm-threshold tests are finite exact rational positive-semidefinite tests.

Proof. Put $e = (k-1)d/K$ and $\gamma = 4ke \leq 1/2$. For a target tuple form $E'_j = (1-\gamma)E_j + \gamma I_d/k$. Each E'_j has margin $2e$ from both spectral faces in (9.1), and $\|E'_j - E_j\|_{\text{op}} \leq 2e$. Round the independent real coordinates of the first $k-1$ matrices to nearest multiples of $1/K$, with a fixed tie rule and Hermitian conjugacy. The row-sum bound gives operator error at most d/K for each rounded matrix. Define the last matrix by exact subtraction from I_d ; its error is at most $(k-1)d/K = e$. Thus every rounded effect still has margin at least e , their sum is exactly I_d , and their distance from the target is at most $3e \leq 3t/4 < t$. The rounded coordinates occur in the stated enumeration.

Legality is checked on $E_j - I_d/(2k)$ and $3I_d/(2k) - E_j$, for every j . The relation $r(\mathbf{E}, \mathbf{F}) \leq s$ is decided by checking $sI_d \pm (E_j - F_j) \succeq 0$ for every j . Exact Schur elimination, including the zero-pivot case, decides these tests over the Gaussian rationals. Equality is included. $\square$

Theorem 9.4 (A public exact code). *For integers $d, N \geq 1$, $k \geq 2$, and rational $0 < \delta < 1/128$, there is a finite deterministic dictionary with centres in $\mathfrak{P}_{d,k}$ and covering error at most $\delta/2$. Its*

fixed-length index uses

$$(9.10) \quad B = \frac{p}{2} \log_2 N + p \log_2(1/\delta) + O(p \log(k+1))$$

bits, with absolute implied constants. A fixed public decoder into arbitrary legal k -outcome measurements cannot improve this order for worst-case radius δ . The construction is finite and exact; no polynomial running-time or workspace assertion is made.

Proof. Put $n = \lceil \sqrt{N} \rceil$ and $s = \delta/(8kn)$, and enumerate $\mathcal{G}(s)$ from Lemma 9.3. Retain a point precisely when its r -distance from every earlier retained point is strictly greater than s . The list is finite and nonempty. It is an r -cover of the grid at radius s ; together with the grid approximation it covers the body at radius $2s$. Its D_N -covering error is therefore at most $2k\sqrt{N}(2s) \leq \delta/2$.

Distinct retained points have r -distance greater than s . The volume argument of Theorem 9.2 gives dictionary size at most

$$(1 + 2R/s)^p = (1 + 8n/\delta)^p \leq (17\sqrt{N}/\delta)^p.$$

The lower bound follows from the same theorem because the dictionary covers at radius δ , even allowing arbitrary legal centres. Taking the ceiling of the logarithm changes the length by at most one.

Encoder and decoder reconstruct the same ordered dictionary from (d, k, N, δ) ; only its index is transmitted. Unused binary words decode to the fixed uniform tuple. For a supplied rational target, scan the list for the first centre within $2s$, using the exact tests in Lemma 9.3. For a real target supplied with certified rational approximations, search for a grid point at certified strict distance less than s and use its assigned predecessor. Such a point exists at distance at most $3s/4$ by the grid lemma, so progressively finer certified input approximations make the strict test terminate. An unrestricted oracle for arbitrary real coordinates is not part of the encoding model. The covering assertion itself concerns every real target, irrespective of its computational representation. $\square$

The grid has at most $[(K+1)^d(2K+1)^{d(d-1)}]^{k-1}$ candidates before the legality tests. Reconstructing this list and comparing its retained points can cost far more than the transmitted index. The estimate concerns the description of an unknown measurement, not free communication of a target-dependent dictionary.

10. JOINT LEARNING AND FINITE CERTIFICATES FOR A FINITE ALPHABET

We next construct one learner common to all unknown elements of $\mathfrak{P}_{d,k}$. Its training experiments do not use the pair-dependent eigenvectors in the geometric lower bound. The outcome count k is public, as are d, N, δ, η . The upper procedure uses independent one-call probe-reference pairs; coherent collective processing is confined to completed outputs. The converse permits arbitrary coherent adaptive training, with an every-record budget and bounded public stopping.

Theorem 10.1 (Joint finite-outcome learning and coding). *Let $d, N \geq 1$, $k \geq 2$, $0 < \delta \leq 2^{-24}$, and $0 < \eta \leq 1/8$. Write $M_{d,k}^*(N, \delta, \eta)$ for the minimum every-record call budget with worst-case failure at most η under D_N -error δ on $\mathfrak{P}_{d,k}$. There are absolute constants $c, C > 0$ such that*

$$(10.1) \quad c \frac{N}{\delta^2} \left(d^2 + d \log \frac{1}{\eta} \right) \leq M_{d,k}^*(N, \delta, \eta) \leq C \frac{k^3 N}{\delta^2} \left(d^2 + d \log \frac{k}{\eta} \right).$$

For each fixed k , this is a matching order jointly in d, N, δ, η , including $d = 1$:

$$M_{d,k}^*(N, \delta, \eta) = \Theta_k \left(N \delta^{-2} [d^2 + d \log(1/\eta)] \right).$$

The displayed dependence on growing k is not claimed optimal.

For rational accuracy and confidence, the upper procedure has a finite deterministic specification, returns an index with a fixed public decoder into $\mathfrak{P}_{d,k}$, and simultaneously uses

$$(10.2) \quad B = \frac{(k-1)d^2}{2} \log_2 N + (k-1)d^2 \log_2(1/\delta) + O((k-1)d^2 \log(k+1))$$

bits. On its good event the decoded error is at most $5\delta/8$. The same call order is available with finite trusted controls. Classical search, readout storage, known-state preparation, and known-control realization are separate resources.

Proof. We first give the constructive upper bound for rational parameters. Set $n = \lceil \sqrt{N} \rceil$ and

$$(10.3) \quad \epsilon = \frac{\delta}{16kn}.$$

For each j consider the binary effect $F_j = I_d/4 + E_j/2$. It belongs to $[I_d/4, 3I_d/4]$. One call to $\mathcal{M}_{\mathbf{E}}$ simulates a call to $\mathcal{M}_{F_j}$: conditional on the original label x , return bit one with probability $3/4$ when $x = j$ and with probability $1/4$ otherwise. This is an equality of channels, including their action on a retained reference, because the bit-one effect is

$$\frac{3}{4}E_j + \frac{1}{4} \sum_{x \neq j} E_x = \frac{1}{4}I_d + \frac{1}{2}E_j.$$

Only public classical randomness is added. Each simulated call uses exactly one call to the unknown device.

Apply Theorem S15.3 to each F_j at future horizon one, accuracy ϵ , and failure η/k , using fresh records for different j . Its output $\hat{F}_j$ is Gaussian rational and legal. The binary identity $d_1(F_j, \hat{F}_j) = 2\|F_j - \hat{F}_j\|_{\text{op}}$ implies that, with probability at least $1 - \eta$, all the Hermitian matrices $B_j = 2\hat{F}_j - I_d/2$ obey

$$(10.4) \quad \max_j \|B_j - E_j\|_{\text{op}} \leq \epsilon.$$

Only a union bound is needed; no independence assertion concerning selected quantum readout outcomes is used.

The matrices B_j need not sum to I_d . Reconstruct the finite legal grid $\mathcal{G}(\epsilon/2)$ and choose its first tuple $\mathbf{G}$ satisfying

$$\|G_j - B_j\|_{\text{op}} \leq 2\epsilon \quad \text{for every } j.$$

Each comparison is an exact rational positive-semidefinite test. On (10.4), such a tuple exists by Lemma 9.3; it has $r(\mathbf{G}, \mathbf{E}) \leq 3\epsilon$. On every other record, if the finite search finds no tuple, return $\mathbf{U}$. Thus every record has a legal output and the same predetermined training budget. The geometric upper bound gives

$$D_N(\mathbf{E}, \mathbf{G}) \leq 6k\sqrt{N}\epsilon \leq 3\delta/8.$$

Encode $\mathbf{G}$ with the dictionary of Theorem 9.4 at parameter $\delta/2$. Its additional error is at most $\delta/4$, so the decoded error is at most $5\delta/8$. It uses no further device calls and has length (10.2).

There are k binary learners, each using at most $C_0\epsilon^{-2}[d^2 + d\log(k/\eta)]$ calls. Since $n^2 \leq 4N$, this proves the upper bound. The inherited rational synthesis and finite-control theorem applies to the simulated binary channels as to any channels of the same form. The public $1/4, 3/4$ relabeling is exact classical control. The error guarantee used in (10.4) is the inherited actual finite-control guarantee, not an idealized readout subsequently assumed exact. The construction terminates in finite classical time, with no assertion that this time is polynomial. For nonrational error parameters, choose rational $\delta' \in [\delta/2, \delta]$ and $\eta' \in [\eta/2, \eta]$ to obtain the stated existence bound; these choices change only absolute constants.

For the lower bound let $m = \lfloor k/2 \rfloor$ and $q = 2m/k \geq 2/3$. Embed any binary interior effect $F \in [I_d/4, 3I_d/4]$ by the tuple

$$(10.5) \quad E_j(F) = \begin{cases} 2F/k, & 1 \leq j \leq m, \\ 2(I_d - F)/k, & m < j \leq 2m, \\ I_d/k, & j = k = 2m + 1. \end{cases}$$

This belongs to $\mathfrak{P}_{d,k}$. A query to this tuple is simulated using one binary F query: with probability q , retain the binary outcome and select a uniform label from its corresponding group of m labels; otherwise erase the bit and return the extra label. In the odd case the erased reference state is the sum of its two subnormalized binary states, as required by the effect I_d/k . Thus the simulation is valid for entangled inputs and adaptive controls, not merely for outcome probabilities on pure states. For even k there is no erasure.

Suppose a k -outcome learner returns any legal tuple $\hat{E}$ within D_N -error δ . Coarse-graining the first m labels gives a binary effect $H = \sum_{j \leq m} \hat{E}_j$ satisfying $d_N(qF, H) \leq \delta$. This follows by executing the same coarse-graining inside any binary tester of the two induced channels. The Bernoulli lower bound in Theorem 9.1, applied to these two binary effects, implies

$$\|H - qF\|_{\text{op}} \leq 128\delta/\sqrt{N},$$

since $\delta < 1/128$. Spectrally clip H/q to $[I_d/4, 3I_d/4]$, obtaining $\hat{F}$. Weyl's inequality bounds the clipping displacement by $\|H/q - F\|_{\text{op}}$; the triangle inequality therefore gives

$$\|\hat{F} - F\|_{\text{op}} \leq \frac{256}{q} \frac{\delta}{\sqrt{N}} \leq \frac{384\delta}{\sqrt{N}}.$$

No operator-Lipschitz claim about spectral clipping is used. The last accuracy is at most 2^{-14} under the stated cap. The proof of Corollary S12.4 gives the binary operator-loss lower bound already on this interior, with arbitrary coherent adaptive training. Applying it to the simulation proves the left inequality of (10.1), including the contribution $d \log(1/\eta)$ and the scalar case. The same simulation preserves every-record budgets and public stopping. $\square$

The length in (10.2) has the optimal leading order also among learners with a fixed-length, fixed public decoder. Indeed, if failure is less than one at every target, at least one word must decode inside that target's loss ball. The decoder's image is therefore a covering of the entire target family. The lower bound in Theorem 9.2 applies regardless of the number of training calls. This is separate from the statistical query lower bound. Shared-randomness expected-prefix conventions are different and are not part of this assertion.

10.1. Finite uniform-risk certificates. The certification mechanism also survives a finite outcome alphabet. For an ideal one-call maximally entangled acquisition, its normalized classical-quantum record is

$$(10.6) \quad \Omega(\mathbf{E}) = \frac{1}{d} \bigoplus_{j=1}^k E_j^\top.$$

The classical labels remain ordered. A collective readout after M independent acquisitions can be described by one POVM $(A_{c,x})_c$ on dimension d^M for each classical string $x \in \{1, \dots, k\}^M$. Here $A_{c,x} \succeq 0$ and $\sum_c A_{c,x} = I_{d^M}$. Its output label c is decoded by a fixed finite family $\mathbf{C}_c$ of legal k -outcome tuples.

Theorem 10.2 (Finite-alphabet certificate transfer). *Let $\mathcal{G} \subset \mathfrak{P}_{d,k}$ be a complete finite r_0 -net in the tuple norm r . Suppose a fixed readout satisfies*

$$(10.7) \quad \sum_{c: r(\mathbf{G}, \mathbf{C}_c) \leq a} \sum_{x \in \{1, \dots, k\}^M} \text{tr} \left[A_{c,x} d^{-M} \bigotimes_{i=1}^M G_{x_i}^\top \right] \geq 1 - \alpha \quad (\mathbf{G} \in \mathcal{G}).$$

Then every real $\mathbf{E} \in \mathfrak{P}_{d,k}$ satisfies

$$(10.8) \quad \mathbb{P}_{\mathbf{E}}\{r(\mathbf{E}, \mathbf{C}_J) > a + r_0\} \leq \alpha + \frac{Mkr_0}{2}.$$

If all decoding centres are in $\mathfrak{P}_{d,k}$, the same failure bound holds for D_N -error greater than $2k\sqrt{N}(a + r_0)$. For Gaussian-rational data, rational a, r_0, α , and the net of Lemma 9.3, the hypotheses admit a finite exact verification. A proper prefix of the net or of the classical string list is not a certificate.

Proof. The block-diagonal formula (10.6) gives

$$\|\Omega(\mathbf{E}) - \Omega(\mathbf{G})\|_1 = d^{-1} \sum_j \|E_j - G_j\|_1 \leq kr(\mathbf{E}, \mathbf{G}).$$

Tensor telescoping between normalized states gives

$$(10.9) \quad \|\Omega(\mathbf{E})^{\otimes M} - \Omega(\mathbf{G})^{\otimes M}\|_1 \leq Mkr(\mathbf{E}, \mathbf{G}).$$

Apply the same readout to both states. Its index laws differ in total variation by at most $Mkr_0/2$ when $r(\mathbf{E}, \mathbf{G}) \leq r_0$. Choose such a $\mathbf{G}$. The fixed set of labels $\{c : r(\mathbf{G}, \mathbf{C}_c) \leq a\}$ has probability at least $1 - \alpha$ at $\mathbf{G}$, by (10.7); every label in this set has $r(\mathbf{E}, \mathbf{C}_c) \leq a + r_0$. Transferring this one event proves (10.8). There is no assumption that the success-label sets vary continuously with the unknown target.

The geometric upper comparison proves the assertion about D_N . For exact verification, reconstruct the whole legal net and every classical string. All traces in (10.7) are Gaussian-rational expressions with rational real values. Check complete POVM positivity and normalization at every string, use $sI_d \pm (G_j - C_{c,j}) \succeq 0$ to determine the good labels including equality, and check each rational inequality. These are finite computations. Their soundness on the real parameter body is supplied by (10.9), not by a numerical sampling argument. $\square$

The certificate describes the ideal specified rational readout. When implemented controls have a separately certified output-law error ζ , the right side of (10.8) increases by at most ζ . Such an error must be budgeted, not silently identified with zero. The finite-control guarantee in Theorem 10.1 instead uses the already budgeted binary procedures. The exact certificate transfer and the actual-control learning guarantee thus have explicitly distinct hypotheses.

11. SEARCH-FREE LEGALIZATION OF BALANCED ESTIMATES

The exact public dictionary and the statistical legalization step have different purposes. A dictionary exploits global packing to reduce transmitted length. Legalizing a collection of component estimates need not reconstruct such a dictionary.

Theorem 11.1 (Affine repair with a uniform error bound). *Let $B = (B_1, \dots, B_k)$ be Hermitian matrices and $a > 0$. Define*

$$(11.1) \quad A_j = B_j + \frac{I_d - \sum_i B_i}{k},$$

$$(11.2) \quad L_a(B)_j = \frac{A_j + 4aI_d}{1 + 4ka}.$$

Suppose $E \in \mathfrak{P}_{d,k}$, the balanced interior from (9.1), and $\max_j \|B_j - E_j\|_{\text{op}} \leq a$. Then $L_a(B) \in \mathfrak{P}_{d,k}$ and

$$(11.3) \quad \max_j \|L_a(B)_j - E_j\|_{\text{op}} \leq \frac{4a}{1+4ka} \leq 4a.$$

For rational input and rational a , this is a rational map. Testing its balanced legality and returning the uniform tuple upon failure defines a total legal procedure on every Hermitian input record. It uses $O(kd^2)$ rational arithmetic operations apart from $2k$ exact positive-semidefinite tests of order d , and polynomial bit complexity. There is no grid enumeration.

Proof. Let $\Delta_j = B_j - E_j$. Then $A_j - E_j = \Delta_j - k^{-1} \sum_i \Delta_i$, whose norm is at most $2a$, and $\sum_j A_j = I_d$. Consequently

$$(1/(2k) - 2a)I_d \preceq A_j \preceq (3/(2k) + 2a)I_d.$$

Adding $4aI_d$ and dividing by $1+4ka$ gives exactly the two balanced spectral inequalities. The sum of the repaired effects is I_d . Finally,

$$L_a(B)_j - E_j = \frac{A_j - E_j + 4ka(I_d/k - E_j)}{1+4ka}.$$

The balanced hypothesis gives $\|I_d/k - E_j\|_{\text{op}} \leq 1/(2k)$, proving (11.3). Rationality and the operation count follow from the formula. The exact legality test makes no good-event assumption; if it fails the uniform tuple is an explicitly specified legal output. On the good event it never fails. $\square$

Corollary 11.2 (A common learner with polynomial legalization). *Under the range and access conventions of Theorem 10.1, replace its tuple-grid legalization by the guarded map in Theorem 11.1. Taking*

$$a = \frac{\delta}{32k\lceil\sqrt{N}\rceil}$$

and allocating failure η/k to each fresh binary component learner gives, on every record,

$$M \leq Ck^3N\delta^{-2}[d^2 + d\log(k/\eta)]$$

for an absolute C , with future-use error at most $\delta/4$ on a common event of probability at least $1 - \eta$. Encoding with the public dictionary at parameter $\delta/2$ gives error at most $\delta/2$ on the same event, without further calls, and retains the optimal fixed- k fixed-length description order. Only the legalization cost is made polynomial; neither dictionary construction nor general collective-readout synthesis is asserted to be efficient.

Proof. Use precisely the reference-preserving binary simulation and actual finite-control component learners from Theorem 10.1, at the displayed smaller accuracy. Its conditional good-event argument gives component error at most a with total failure at most η . Affine repair then gives tuple error at most $4a$. The balanced metric upper bounds the future-use error by $8k\sqrt{N}a \leq \delta/4$. The dictionary adds at most $\delta/4$. Fresh calls and all finite-control budgets are charged exactly as in the component theorem; decreasing the component accuracy by a factor two changes its call bound only by an absolute factor. The guarded map is total even off the good event, so the recordwise call bound and public stopping convention are unchanged. $\square$

12. RELATION TO CHANNEL METROLOGY AND MEASUREMENT INFORMATION

12.1. Support spans, channel estimation and finite pairs. For the measurement Kraus operators $L_{j,a} = |j\rangle\langle a|\sqrt{E_j}$, the real Hermitian Kraus-product span is exactly $\mathcal{W}_E$. For the orbit $L_{j,a}(t) = L_{j,a}e^{-itG}$ the effective generator is G . Zhou–Jiang [40, Theorems 1–3 and Section V] establish the Kraus-span criterion for the linear or quadratic asymptotic channel Fisher-information

rate and give error-correction attainability. We do not claim a new metrological dichotomy or a new correction principle. Theorems 4.1 and 5.1 identify its complete support-kernel form and give finite-angle trace-distance inequalities with an explicit $4mt^2\|G\|_{\text{op}}^2$ remainder. Theorem 6.2 adds a normalized first-order realization for general two-sided C^2 effect curves; its $\sqrt{N}|t| + O(Nt^2)$ bound handles rank changes without a smooth exact Kraus hypothesis. Theorem 7.1 separately budgets certified control errors. Neither constitutes a new claim to the underlying metrological exponents.

Sieniawski–Demkowicz–Dobrzański [26, Section IV, equations (16)–(23)] integrate channel Fisher-information bounds to bound adaptive discrimination and compare metrological scaling with perfect finite discrimination. The same horizontal method is a principal antecedent of (1.1). Our support-kernel identity is stated in measurement coordinates and our lower (5.9) comes from an actual corrected finite-angle channel. It does not claim the exact optimal tester or perfect discrimination.

12.2. Fisher frames and covariance embeddings. Saini–Kiukas–Burgarth–Gilchrist [42, Theorem 1] compare classical and quantum Fisher information for a varying full-rank *state* measured by one fixed informationally complete POVM, using a state-dependent frame operator. Here the varying object is the measurement, the tangent has coupled normalization, and the loss permits N adaptive uses with a reference. In the rank-one case our support span equals the effect span, so informational completeness excludes the linear orbit regime by Corollary 5.4. That intersection does not identify the two quadratic forms or their resource models.

Mayer–Yun [43, Theorem 3.6, Definition 3.7 and Remark 4.4] use a kernel embedding of operator-valued measures, a kernel-dependent discrepancy, and a tomographic Gram operator whose nullspace records unobserved state directions. Our support span contains entire matrix blocks $P_j\mathcal{H}_dP_j$, rather than one observable per effect; this distinction disappears for rank-one effects but not in general. Theorem 4.1 concerns a coupled measurement tangent, and (1.1) concerns adaptive finite-use loss. Neither kernel embedding nor injectivity alone supplies those statements.

12.3. Known symmetry and tomography primitives. The full primary text of Yoshida–Okigami–Posta–Grinko [44] distinguishes known-symmetry estimation from known-pair discrimination. Theorem 7 gives invariant-state copy complexity $\Theta((D_{\text{st}} - 1 + \log(1/\eta))/\epsilon^2)$, where $D_{\text{st}} = \sum_{\lambda} m_{\lambda}^2$; $D_{\text{st}} = 1$ is a known singleton. Corollary 9 gives parallel covariant-channel bounds

$$O((D'_G + K_G \log(1/\eta))/\epsilon_{\diamond}^2), \quad O((C'_G + \log(1/\eta))/\epsilon_{\text{tr}}^2),$$

in one-use halved diamond distance and normalized-Choi trace distance, respectively, for $0 < \epsilon \leq 1$, $0 < \eta \leq 1/2$. Here $D_G = \sum_{\lambda} m_{\lambda} M_{\lambda}$ counts Choi commutant parameters before trace preservation, $K_G = \sum_{\lambda} m_{\lambda}$, $D'_G = D_G + K_G \log(4L)$, and $C'_G = d_{\mathcal{I}}^{-1}(\sum_{\lambda} m_{\lambda} \sqrt{d_{\lambda} M_{\lambda}})^2$; L counts occupied blocks. Theorem 13 does not give a general matching inverse-square diamond lower: its bound is $\Omega_{\xi}(D_G/\epsilon_{\diamond}^{\xi/2})$ under its affine-dimension hypothesis.

For permutation covariance on ℓ systems of fixed local dimension q , Theorems 12 and 15 match the Choi order $\ell^{q^4-q^2}\epsilon_{\text{tr}}^{-2}$ at constant confidence. Theorem 18 approximates the Hayashi measurement in $O((s+q) \text{polylog}(s, q, \tau^{-1}, \zeta^{-1}))$ gates, with coupled output error at most τ except with probability ζ . This is not a diamond-error certificate for our recovery. Known-symmetry estimation, its parameter dimensions, and that specialized gate construction do not themselves give our support kernel or finite-pair curve theorem. Nor do our theorems claim faster learning on those symmetry classes.

For ordinary tomography the main-text substitutions remain explicit. Mele–Bittel [5, Theorem III.3] give the sufficient constant-confidence order $O(dkr\epsilon^{-2})$ in halved diamond loss, with $r \leq dk$ for measurement channels. Zambrano–Ramos-Calderer–Kueng [29, Theorem 2] give the

sufficient single-copy worst-input outcome-TV bound

$$m \geq \frac{8[d^2 + \epsilon(d^2 + 1)/6]}{\epsilon^2} \log \frac{2^{k+1}9^{2d}}{\eta}.$$

Either loss bounds the component norm error by ϵ after output dephasing where needed. A legal estimator need not be balanced: apply Theorem 11.1 before using the interior metric. The resulting future loss is at most $8k\sqrt{N}\epsilon$. With $\epsilon = \delta/(8k\sqrt{N})$ the respective sufficient orders are

$$O(d^2k^4N\delta^{-2}), \quad O\left(k^2N\delta^{-2}d^2[d + k + \log(1/\eta)]\right).$$

These are upper-bound substitutions, not optimality claims about the cited estimators. Our component procedure credits the existing binary operator-norm primitive and uses a different confidence allocation. The full calculations and confidence restrictions are retained in the supplement, Section S3.

12.4. What remains beyond the theorems. The orbit theorem and Theorem 6.2 provide matching scales on each fixed two-sided smooth curve with nonzero first derivative, including second-order rank openings. Their constants need not be uniform across curves. Arbitrary-pair midpoint equivalence and full-body multi-outcome entropy are not consequences of these local statements. Growing- k minimax sharpness, polynomial entropy-optimal dictionaries, and general readout synthesis remain separate mathematical questions. Independent expert priority assessment is not inferred from the absence of an identical statement in this targeted comparison.

REFERENCES

- [1] J. Watrous, *The Theory of Quantum Information*, Cambridge University Press, 2018. Theorems 2.22 and 2.26 (Choi criteria), Corollary 3.40 (trace-norm contraction), and Section 3.3, equation (3.306) (hybrid comparison).
- [2] T. Surawy-Stepney, J. Kahn, R. Kueng and M. Guță, Projected least-squares quantum process tomography, *Quantum* **6** (2022), 844; arXiv:2107.01060v2, 18 October 2022.
- [3] G. Gutoski, On a measure of distance for quantum strategies, *J. Math. Phys.* **53** (2012), 032202; arXiv:1008.4636v4, 14 March 2012.
- [4] S. G. Naik, M. Zartab, N. Gisin and M. Banik, No-Go Theorem for Generic Simulation of Qubit Channels with Finite Classical Resources, arXiv:2501.15807v2 (16 July 2025), Theorems 2 and 5.
- [5] A. A. Mele and L. Bittel, Optimal learning of quantum channels in diamond distance, arXiv:2512.10214v3 (2026), Theorem I.1, Lemmas III.7–III.8, Corollary III.9 and Sections III.1.1, IV.1.1.
- [6] A. Oufkir and F. Girardi, Improved Lower Bounds for Learning Quantum Channels in Diamond Distance, arXiv:2601.04180v4 (2026).
- [7] R. Demkowicz-Dobrzański, J. Kołodyński and M. Guță, The elusive Heisenberg limit in quantum-enhanced metrology, *Nature Communications* **3** (2012), 1063. DOI: 10.1038/ncomms2067; arXiv:1201.3940v2, equations (4)–(6) (classical simulation), equations (10)–(13) (channel extension), and the classical-simulation argument in Methods.
- [8] Y. Yang, G. Bai, G. Chiribella and M. Hayashi, Compression for quantum population coding, *IEEE Transactions on Information Theory* **64** (2018), no. 7, doi:10.1109/TIT.2017.2788407; arXiv:1701.03372v8.
- [9] F. Salek, M. Hayashi and A. Winter, Usefulness of adaptive strategies in asymptotic quantum channel discrimination, *Phys. Rev. A* **105** (2022), 022419; arXiv:2011.06569v3 (17 March 2024).
- [10] T. Cooney, M. Mosonyi and M. M. Wilde, Strong converse exponents for a quantum channel discrimination problem and quantum-feedback-assisted communication, arXiv:1408.3373v3 (26 February 2016), p. 5 and Sections 4.1–4.2.
- [11] A. Acín, Statistical distinguishability between unitary operations, *Phys. Rev. Lett.* **87** (2001), 177901. DOI: 10.1103/PhysRevLett.87.177901.
- [12] R. Duan, Y. Feng and M. Ying, Entanglement is not necessary for perfect discrimination between unitary operations, *Phys. Rev. Lett.* **98** (2007), 100503. DOI: 10.1103/PhysRevLett.98.100503.
- [13] S. Das and M. M. Wilde, Quantum reading capacity: General definition and bounds, *IEEE Trans. Inform. Theory* **65** (2019), 7566–7583; arXiv:1703.03706v3; doi:10.1109/TIT.2019.2929925.
- [14] M. M. Wilde, M. Berta, C. Hirche and E. Kaur, Amortized channel divergence for asymptotic quantum channel discrimination, *Lett. Math. Phys.* **110** (2020), 2277–2336; arXiv:1808.01498v2; doi:10.1007/s11005-020-01297-7.

- [15] X. Wang and M. M. Wilde, Resource theory of asymmetric distinguishability for quantum channels, *Phys. Rev. Research* **1** (2019), 033169; arXiv:1907.06306v2; doi:10.1103/PhysRevResearch.1.033169.
- [16] Z. Puchała, Ł. Paweła, A. Krawiec, R. Kukulski and M. Oszmaniec, Multiple-shot and unambiguous discrimination of von Neumann measurements, *Quantum* **5** (2021), 425; arXiv:1810.05122v4; doi:10.22331/q-2021-04-06-425.
- [17] M. Sedláč and M. Ziman, Optimal single-shot strategies for discrimination of quantum measurements, *Phys. Rev. A* **90** (2014), 052312; arXiv:1408.0934; doi:10.1103/PhysRevA.90.052312. Section VI, Example 4, equation (37).
- [18] Z. Puchała, Ł. Paweła, A. Krawiec and R. Kukulski, Strategies for optimal single-shot discrimination of quantum measurements, *Phys. Rev. A* **98** (2018), 042103; arXiv:1804.05856; doi:10.1103/PhysRevA.98.042103. Theorem 1.
- [19] J. Fiurášek and M. Mičuda, Optimal two-copy discrimination of quantum measurements, *Phys. Rev. A* **80** (2009), 042312; arXiv:0909.2940; doi:10.1103/PhysRevA.80.042312.
- [20] A. Krawiec, Ł. Paweła and Z. Puchała, Discrimination of POVMs with rank-one effects, *Quantum Information Processing* **19** (2020), 428; arXiv:2002.05452v1; doi:10.1007/s11128-020-02883-3.
- [21] C. Datta, T. Biswas, D. Saha and R. Augusiak, Perfect discrimination of quantum measurements using entangled systems, *New Journal of Physics* **23** (2021), 043021; arXiv:2012.07069v2; doi:10.1088/1367-2630/abecaf.
- [22] A. Fujiwara and H. Imai, A fibre bundle over manifolds of quantum channels and its application to quantum statistics, *J. Phys. A: Math. Theor.* **41** (2008), 255304; doi:10.1088/1751-8113/41/25/255304. Theorems 4 and 5.
- [23] R. Demkowicz-Dobrzański and L. Maccone, Using entanglement against noise in quantum metrology, *Phys. Rev. Lett.* **113** (2014), 250801; arXiv:1407.2934v3; doi:10.1103/PhysRevLett.113.250801.
- [24] S. Kurdziałek, W. Górecki, F. Albarelli and R. Demkowicz-Dobrzański, Using adaptiveness and causal superpositions against noise in quantum metrology, *Phys. Rev. Lett.* **131** (2023), 090801; arXiv:2212.08106v2; doi:10.1103/PhysRevLett.131.090801.
- [25] H. Yuan and C.-H. F. Fung, Fidelity and Fisher information on quantum channels, arXiv:1506.00819v3 (2017), Section III, equation (26).
- [26] S. Sieniański and R. Demkowicz-Dobrzański, Adaptive quantum channel discrimination using methods of quantum metrology, *New Journal of Physics* **28** (2026), 024502; arXiv:2510.15506v3; doi:10.1088/1367-2630/ae3edd.
- [27] J. Dittmann, Explicit formulae for the Bures metric, *J. Phys. A: Math. Gen.* **32** (1999), 2663–2670; doi:10.1088/0305-4470/32/14/007; arXiv:quant-ph/9808044, entitled *Note on explicit formulae for the Bures metric*.
- [28] H.-J. Sommers and K. Życzkowski, Bures volume of the set of mixed quantum states, *J. Phys. A: Math. Gen.* **36** (2003), 10083–10100; arXiv:quant-ph/0304041v3; doi:10.1088/0305-4470/36/39/308.
- [29] L. Zambrano, S. Ramos-Calderer and R. Kueng, Fast quantum measurement tomography with optimal error bounds, *Quantum* **10** (2026), 2162; arXiv:2507.04500v3; doi:10.22331/q-2026-07-15-2162.
- [30] A. Bisio, G. M. D’Ariano, P. Perinotti and M. Sedláč, Quantum learning algorithms for quantum measurements, *Physics Letters A* **375** (2011), 3425–3434; arXiv:1103.0480v2; doi:10.1016/j.physleta.2011.08.002.
- [31] B. L. Higgins, D. W. Berry, S. D. Bartlett, M. W. Mitchell, H. M. Wiseman and G. J. Pryde, Demonstrating Heisenberg-limited unambiguous phase estimation without adaptive measurements, *New Journal of Physics* **11** (2009), 073023; arXiv:0809.3308v4; doi:10.1088/1367-2630/11/7/073023.
- [32] S. Kimmel, G. H. Low and T. J. Yoder, Robust calibration of a universal single-qubit gate set via robust phase estimation, *Phys. Rev. A* **92** (2015), 062315; doi:10.1103/PhysRevA.92.062315; erratum, *Phys. Rev. A* **104** (2021), 069901, doi:10.1103/PhysRevA.104.069901; arXiv:1502.02677v3.
- [33] P. Hyllus, W. Laskowski, R. Krischek, C. Schwemmer, W. Wieczorek, H. Weinfurter, L. Pezzé and A. Smerzi, Fisher information and multiparticle entanglement, *Phys. Rev. A* **85** (2012), 022321; arXiv:1006.4366v3; doi:10.1103/PhysRevA.85.022321. Section III.A, Observation 1, equation (9).
- [34] G. Tóth, Multipartite entanglement and high-precision metrology, *Phys. Rev. A* **85** (2012), 022322; arXiv:1006.4368v3; doi:10.1103/PhysRevA.85.022322. Observation 3, equation (6).
- [35] K. Chen, Z. Zhang and N. Yu, Optimal lower bound for quantum channel tomography in away-from-boundary regime, arXiv:2601.10683v1 (15 January 2026), Theorems 1.1, 3.2 and 4.2.
- [36] S. Chen, J. Li and A. Liu, An optimal tradeoff between entanglement and copy complexity for state tomography, arXiv:2402.16353v1 (26 February 2024).
- [37] A. Nayak and X. Zhou, Optimal Low-Rank Quantum State Tomography with Bounded-Sample Joint Measurements, arXiv:2609.10514v1 (9 September 2026), Theorems 1.1–1.2.
- [38] S. Basu, R. Pollack and M.-F. Roy, *Algorithms in Real Algebraic Geometry*, second edition, Algorithms and Computation in Mathematics, vol. 10, Springer, Berlin, Heidelberg, 2006; doi:10.1007/3-540-33099-2. Theorems 2.77 and 2.80 and Chapter 14.

- [39] A. Pelecanos, J. Spilecki, E. Tang and J. Wright, Mixed state tomography reduces to pure state tomography, arXiv:2511.15806v1 (19 November 2025). Theorems 1.1–1.3 and 4.2, Section 3.3, Figures 8–9.
- [40] S. Zhou and L. Jiang, Asymptotic theory of quantum channel estimation, *PRX Quantum* **2** (2021), 010343; arXiv:2003.10559v3; doi:10.1103/PRXQuantum.2.010343.
- [41] E. Knill and R. Laflamme, A theory of quantum error-correcting codes, *Phys. Rev. A* **55** (1997), 900–911; arXiv:quant-ph/9604034; doi:10.1103/PhysRevA.55.900.
- [42] R. Saini, J. Kiukas, D. Burgarth and A. Gilchrist, Characterizing Fisher information of quantum measurement, arXiv:2512.15428v1 (2025).
- [43] P. N. Mayer and H. Yun, Kernel embedding for operator-valued measures and its application to quantum tomography, arXiv:2605.25146v1 (2026).
- [44] S. Yoshida, K. Okigami, P. M. Posta and D. Grinko, Optimal learning of covariant quantum states and channels, arXiv:2609.39280v1 (2026), Theorem 7, Corollary 9 and Theorems 12–13, 15 and 18; full text consulted 5 October 2026.